Lecture 2: Detailed Solutions

Real book exercises — Economics of Organizations and Incentives (MPE Insper)

Solutions to the 22 exercises of `Listas/pratica-lecture-02.md`

November 2026

These are detailed solutions to the 22 real book exercises of `Listas/pratica-lecture-02.md`, in the list's order, which runs from easy to hard. Each heading shows the item's number in the list, its level and its difficulty from 1 (apply a definition) to 5 (graduate level). Each statement is printed verbatim from the list right before its solution. Every solution has the same parts: (a) what is asked, in one line; (b) the method, with the book section and page and the lecture slide (printed slide numbers of `Aula/Lecture02.pdf`); (c) the solution, every step shown; (d) the economic interpretation; (e) the verification, by the Python check scripts and against the book's worked answer or the MWG solutions manual where one exists; (f) common mistakes. Ten solutions (items 1, 2, 6, 8, 12, 14, 15, 16, 17, 21) were written for the earlier list `lista-reais-lectures-01-05` and are input from `../lista-reais-lectures-01-05-solutions/`; they are verified by `check_A2.py` there. The twelve others are new and are verified by `check_L2.py` in this folder. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992), Mas-Colell, Whinston & Green (1995) (MWG). In bimatrices player 1 picks rows and the first payoff is player 1's; underlined payoffs are best replies.

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Lecture 2 — Static games of complete information II: Nash equilibrium, Cournot, Bertrand and mixed strategies

Item 1: Muñoz-Garcia Ch. 2, Exercise 2

Foundations · difficulty 1/5

Statement (verbatim from the list, item 1)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 2, Exercise 2 "Battle of the Sexes", p. 41 [PDF p. 54]. Answer in the source: yes, a worked answer follows the exercise. Reproduced verbatim (statement only; the book's worked answer follows it).

> A husband and a wife are leaving work, and do not remember which event they are attending to tonight. Both of them, however, remember that last night's argument was about either attending to the football game (the most preferred event for the husband) or the opera (the most preferred event for the wife). To make matters worse, their cell phones broke, so they cannot call each other to confirm which event they are attending to. As a consequence, they must simultaneously and independently decide whether to attend to the football game or the opera.
>
> The payoff matrix in Fig. 2.4 describes the preference of the husband (wife) for the football game (opera, respectively), but also indicates that both players prefer to be together rather than being alone (even if they are alone at their most preferred event). Find the set of Nash Equilibria of this game.

Housband [sic] *Wife*	*Football*	*Opera*
Football	3, 1	0, 0
Opera	0, 0	1, 3

Fig. 2.4 Battle of the sexes game (Normal-form representation)

(a) What is asked. Find the set of Nash equilibria of the Battle of the Sexes in Fig. 2.4.

(b) Method. Pure-strategy Nash equilibrium by best-response marking (Tadelis §5.1, pp. 80–81; Lecture 2, slides 10 and 14; the slides' own Battle of the Sexes, slide 9). Muñoz-Garcia's Chapter 2 covers pure strategies only, so the book's "set of Nash Equilibria" is the pure set; for completeness the mixed equilibrium is added by the principle of indifference (Tadelis §6.2 and Prop. 6.1, pp. 107–108; Lecture 2, slides 44–50, where the same game is solved in mixed strategies).

(c) Solution. **Notation.** Husband H (rows), wife W (columns); F = Football, O = Opera.

Best responses, underlined:

$H \backslash W$	F	O
F	$\underline{3}, \underline{1}$	$0, 0$
O	$0, 0$	$\underline{1}, \underline{3}$

- Husband: against F, $3 > 0$, so $BR_H(F) = \{F\}$; against O, $1 > 0$, so $BR_H(O) = \{O\}$.
- Wife: against F, $1 > 0$, so $BR_W(F) = \{F\}$; against O, $3 > 0$, so $BR_W(O) = \{O\}$.

Each player's best reply is to go where the other goes.

Pure equilibria, cell by cell:

- (F, F): $F \in BR_H(F)$ and $F \in BR_W(F)$. NE, payoffs $(3, 1)$.
- (O, O): NE by the same check, payoffs $(1, 3)$.
- (F, O): the husband would switch to O $(1 > 0)$. Not NE.
- (O, F): the husband would switch to F $(3 > 0)$. Not NE.

psNE $= \{(F, F), (O, O)\}$.

The mixed equilibrium. Let $p = \Pr(H \text{ plays } F)$ and $q = \Pr(W \text{ plays } F)$.

- Husband indifferent: $EU_H(F) = 3q$, $EU_H(O) = 1 \cdot (1 - q)$, so $3q = 1 - q$ and $q^* = \frac{1}{4}$.
- Wife indifferent: $EU_W(F) = 1 \cdot p$, $EU_W(O) = 3(1 - p)$, so $p = 3 - 3p$ and $p^* = \frac{3}{4}$.

At $(p^*, q^*) = (\frac{3}{4}, \frac{1}{4})$ each player is indifferent between both pure strategies, so each mix is a best reply to the other (Lecture 2, slide 44). Expected payoffs: $EU_H = 3q^* = \frac{3}{4}$ and $EU_W = p^* = \frac{3}{4}$. The full set of Nash equilibria is

$$\Big\{(F, F),\ (O, O),\ ((\tfrac{3}{4}F, \tfrac{1}{4}O),\ (\tfrac{1}{4}F, \tfrac{3}{4}O))\Big\}.$$

There are no others: $BR_H(q)$ is $p = 1$ for $q > \frac{1}{4}$, $p = 0$ for $q < \frac{1}{4}$ and all of $[0, 1]$ at $q = \frac{1}{4}$; $BR_W(p)$ is $q = 1$ for $p > \frac{3}{4}$, $q = 0$ for $p < \frac{3}{4}$ and $[0, 1]$ at $p = \frac{3}{4}$; the two graphs cross only at $(0, 0)$, $(1, 1)$ and $(\frac{3}{4}, \frac{1}{4})$.

(d) Interpretation. A coordination game with conflict: both want to be together, but they disagree on where. Nash equilibrium identifies the two conventions but cannot say which one the couple will use; that needs something outside the game (a focal point, a habit, who argued last night). In the mixed equilibrium each goes to his or her *own* favourite event with probability $\frac{3}{4}$, they end up apart with probability $\frac{3}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{5}{8}$, and each expects $\frac{3}{4}$, less than the 1 either gets in the worse pure equilibrium.

(e) Verification. `check_A2.py` finds by brute force that the pure equilibria are exactly (F, F) and (O, O), solves both indifference conditions ($q^* = \frac{1}{4}$, $p^* = \frac{3}{4}$), checks the payoffs $\frac{3}{4}$ and the miscoordination probability $\frac{5}{8}$, and finds on a grid of mixed profiles that the mutual best responses are exactly $(0, 0)$, $(1, 1)$ and $(\frac{3}{4}, \frac{1}{4})$. *Comparison with the book's answer* (pp. 41–42): it finds the same best responses and psNE $= \{(F, F), (O, O)\}$; it does not report the mixed equilibrium, which belongs to its Chapter 3. The row label "Housband" in Fig. 2.4 is the book's typo.

(f) Common mistakes.

- Swapping the mixes: the husband's indifference pins down the *wife's* probability ($q^* = \frac{1}{4}$), not his own.
- Reading the answer as "the husband goes to football": an equilibrium is a profile, here (F, F) or (O, O), not one player's choice.
- Claiming the mixed equilibrium is "fairer" and therefore better. Both players get $\frac{3}{4}$, less than in either pure equilibrium.

Answer. Best replies: each goes where the other goes. Pure Nash equilibria: (F, F) with payoffs $(3, 1)$ and (O, O) with payoffs $(1, 3)$ (the book's answer). Including mixed strategies, there is one more: the husband plays F with probability $\frac{3}{4}$, the wife plays F with probability $\frac{1}{4}$, each expecting $\frac{3}{4}$.

Item 2: Tadelis Exercise 5.6

Foundations · difficulty 1/5

Statement (verbatim from the list, item 2)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 5.6 "Hawk-Dove", p. 96 [PDF p. 115]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim (with the book's footnote 12).

> **Hawk-Dove:** The following game has been widely used in evolutionary biology to understand how fighting and display strategies by animals could coexist in a population. For a typical Hawk-Dove game there are resources to be gained (e.g., food, mates, territories), denoted as v. Each of two players can choose to be aggressive, as Hawk (H), or compromising, as Dove (D). If both players choose H then they split the resources but lose some payoff from injuries, denoted as k. Assume that $k > \frac{v}{2}$. If both choose D then they split the resources but engage in some display of power that carries a display cost d, with $d < \frac{v}{2}$. Finally, if player i chooses H while j chooses D then i gets all the resources while j leaves with no benefits and no costs.
>
> a. Describe this game in a matrix.
>
> b. Assume that $v = 10$, $k = 6$, and $d = 4$. What outcomes can be supported as pure-strategy Nash equilibria?[12]
>
> *Footnote 12.* In the evolutionary biology literature, the analysis performed is of a very different nature. Instead of considering the Nash equilibrium analysis of a static game, the analysis is a dynamic one in which successful strategies "replicate" in a large population. This analysis is part of a methodology called evolutionary game theory. For more on the subject see Gintis (2000).

(a) What is asked. Write Hawk-Dove as a matrix in (v, k, d), then find its pure-strategy Nash equilibria at $v = 10$, $k = 6$, $d = 4$.

(b) Method. Building a matrix from a story (Lecture 1, slides 17–19); pure-strategy Nash equilibrium (Tadelis §5.1, p. 80; Lecture 2, slide 14), found by marking best responses cell by cell (Lecture 2, slide 10; Tadelis §5.1.1, p. 81). The mixed equilibrium mentioned at the end uses the principle of indifference (Lecture 2, slide 44).

(c) Solution. Notation. $v > 0$: value of the resource; k: injury cost when both fight, $k > v/2$; d: display cost when both display, $d < v/2$ (and $d \geq 0$, a cost). $S_1 = S_2 = \{H, D\}$.

(a) The matrix. Each cell from the story:

- both H: they split v and each loses k: $v/2 - k$ each;
- both D: they split v and each pays d: $v/2 - d$ each;
- one H, one D: the hawk gets all of v, the dove leaves with no benefit and no cost, 0.

$1 \setminus 2$	H	D
H	$\frac{v}{2} - k,\ \frac{v}{2} - k$	$v,\ 0$
D	$0,\ v$	$\frac{v}{2} - d,\ \frac{v}{2} - d$

(b) Pure equilibria at $v = 10$, $k = 6$, $d = 4$. Substituting, $v/2 - k = 5 - 6 = -1$ and $v/2 - d = 5 - 4 = 1$. Best responses underlined:

$1 \setminus 2$	H	D
H	$-1,\ -1$	$\underline{10},\ \underline{0}$
D	$\underline{0},\ \underline{10}$	$1,\ 1$

- Player 1 against H: $u_1(D,H) = 0 > -1 = u_1(H,H)$, so $BR_1(H) = \{D\}$.
- Player 1 against D: $u_1(H,D) = 10 > 1 = u_1(D,D)$, so $BR_1(D) = \{H\}$.
- Player 2, by symmetry: $BR_2(H) = \{D\}$, $BR_2(D) = \{H\}$.

Cell by cell:

- (H,H): each would switch to D ($0 > -1$). Not NE.
- (D,D): each would switch to H ($10 > 1$). Not NE.
- (H,D): $H \in BR_1(D)$ and $D \in BR_2(H)$. NE, payoffs $(10,0)$.
- (D,H): by symmetry, NE, payoffs $(0,10)$.

The same holds for every admissible (v,k,d). Against H, D is strictly better because $0 > v/2 - k \iff k > v/2$. Against D, H is strictly better because $v > v/2 - d \iff d > -v/2$, true for any $d \geq 0$. So the two asymmetric profiles are the only pure equilibria under the book's assumptions; the numbers only make it concrete.

Beyond what is asked: the symmetric mixed equilibrium. Let each play H with probability p. A player is indifferent when

$$\underbrace{p\left(\tfrac{v}{2}-k\right)+(1-p)v}_{EU(H)} = \underbrace{(1-p)\left(\tfrac{v}{2}-d\right)}_{EU(D)} \iff v - p\left(\tfrac{v}{2}+k\right) = \tfrac{v}{2} - d - p\left(\tfrac{v}{2}-d\right) \iff p^* = \frac{v/2+d}{k+d}.$$

At $(10,6,4)$: $p^* = 9/10$, and each expects $EU(D) = (1-0.9)(1) = 0.1$.

(d) Interpretation. Hawk-Dove is an anti-coordination game (the same structure as Chicken, Muñoz-Garcia Ch. 3 Ex. 1): each player wants to do the opposite of the other. The two pure equilibria are asymmetric: one animal takes everything, the other backs off. Nothing in the symmetric game says *which* animal is the hawk, which is why the biology literature looks at the symmetric mixed (population) answer: a share $p^* = 0.9$ of hawks, where hawks and doves do equally well. Fighting is frequent because the prize ($v = 10$) is large relative to the injury loss ($k - v/2 = 1$).

(e) Verification. `check_A2.py` builds the matrix from the story with symbols, checks that $BR(H) = D$ and $BR(D) = H$ for 2,000 random admissible (v,k,d), finds by brute force that the pure NE at $(10,6,4)$ are exactly (H,D) and (D,H), and checks the general formula for p^*, $p^* = 9/10$ and the payoff $1/10$. Tadelis gives no answer.

(f) Common mistakes.

- Giving the dove a negative payoff when it meets a hawk. The story says it leaves "with no benefits and no costs": 0.
- Forgetting that both H and both D split v: the cells are $v/2 - k$ and $v/2 - d$, not $v - k$ and $-d$.
- Answering "(H,D)" only. The game is symmetric, so (D,H) is an equilibrium too.

Answer. (a) (H,H): $(\frac{v}{2}-k, \frac{v}{2}-k)$; (H,D): $(v,0)$; (D,H): $(0,v)$; (D,D): $(\frac{v}{2}-d, \frac{v}{2}-d)$. (b) With $v=10$, $k=6$, $d=4$ the matrix is $\begin{pmatrix} -1,-1 & 10,0 \\ 0,10 & 1,1 \end{pmatrix}$; the pure-strategy Nash equilibria are (H,D) and (D,H), with payoffs $(10,0)$ and $(0,10)$.

Item 3: Tadelis Exercise 5.5

Foundations · difficulty 2/5

Statement (verbatim from the list, item 3)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 5.5 "Public Good Contribution", p. 96 [PDF p. 115]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim.

> **Public Good Contribution:** Three players live in a town, and each can choose to contribute to fund a streetlamp. The value of having the streetlamp is 3 for each player, and the value of not having it is 0. The mayor asks each player to contribute either 1 or nothing. If at least two players contribute then the lamp will be erected. If one player or no players contribute then the lamp will not be erected, in which case any person who contributed will not get his money back.
>
> a. Write out or graph each player's best-response correspondence.
>
> b. What outcomes can be supported as pure-strategy Nash equilibria?

(a) What is asked. Write each player's best-response correspondence in the three-player streetlamp game, then list the outcomes that are pure-strategy Nash equilibria.

(b) Method. Best response and pure-strategy Nash equilibrium (Lecture 2, slides 10 and 14; Tadelis §5.1.1, p. 81). With three players a matrix is awkward, so the payoff of each action is tabulated by the number k of *other* players who contribute; a profile is a Nash equilibrium when every player's action is a best response to the number k the others produce.

(c) Solution. Payoffs. $S_i = \{C, N\}$ (C = contribute 1, N = nothing). The lamp is erected iff at least two players contribute; it is worth 3 to everyone, whether or not they contributed, and a contribution is lost. Fix player i and let $k \in \{0, 1, 2\}$ be the number of the other two who contribute.

k	$v_i(C, k)$	$v_i(N, k)$	best response
0	lamp needs 2: only i pays, no lamp: -1	0	N
1	two contribute, lamp erected: $3 - 1 = 2$	one contributor, no lamp: 0	C
2	lamp erected: $3 - 1 = 2$	lamp erected, i free rides: 3	N

No ties occur, so each best response is unique.

(a) Best-response correspondence. For each player i,

$$BR_i(k) = \begin{cases} N & k = 0, \\ C & k = 1, \\ N & k = 2, \end{cases}$$

where k is the number of contributions by the other two. In words: contribute exactly when you are *pivotal*, that is, when your contribution is the one that makes the second one.

(b) Pure-strategy Nash equilibria. Check the four types of profile (the game is symmetric, so only the number of contributors matters).

- *No one contributes* (N, N, N): each player sees $k = 0$ and $BR = N$. Nash equilibrium.
- *Exactly one contributes*: the contributor sees $k = 0$ and $BR = N$, so he deviates (his -1 becomes 0). Not an equilibrium.
- *Exactly two contribute*, e.g. (C, C, N): each contributor sees $k = 1$ and $BR = C$; the free rider sees $k = 2$ and $BR = N$. Nash equilibrium. There are three such profiles.
- *All three contribute*: each sees $k = 2$ and $BR = N$, so each wants to free ride ($3 > 2$). Not an equilibrium.

Answer. (a) Contribute iff exactly one other player contributes: $BR_i(0) = N$, $BR_i(1) = C$, $BR_i(2) = N$. (b) Four pure-strategy Nash equilibria: nobody contributes and no lamp is built, or exactly two players contribute, which can happen in 3 ways, and the lamp is erected and the third player free rides.

(d) Interpretation. The two kinds of equilibria are the two faces of a public good. When exactly two pay, the lamp exists but the equilibrium is not symmetric and is unfair to the contributors, who earn 2 against the free rider's 3. When nobody pays, a lone contribution is wasted, so a coordination failure is self-enforcing. Full contribution, the efficient outcome (2 each against 0 each), is *not* an equilibrium: free riding on the other two is tempting. This is the same logic as the dilemma of Lecture 2 (slide 18), with the twist that the best response flips with k.

(e) Verification. `check_L2.py` enumerates all 2^3 profiles with the payoff $3 \cdot 1\{\text{at least two contribute}\} - \text{own contribution}$, checks the three table rows and that the equilibrium set is $\{NNN, NCC, CNC, CCN\}$. Tadelis gives no answer.

(f) Common mistakes.

- Treating the lamp as built when one player contributes. The text says one contribution is lost, which is why $v_i(C, 0) = -1$.
- Counting all-contribute as an equilibrium because it is efficient.
- Listing only "two contribute" and forgetting the no-contribution equilibrium.

Item 4: Gibbons Problem 1.3

Foundations · difficulty 2/5

Statement (verbatim from the list, item 4)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §1.1 Problems, Problem 1.3, p. 49 [PDF p. 57]. Answer in the source: none (Gibbons prints no answers). Reproduced verbatim.

> 1.3. Players 1 and 2 are bargaining over how to split one dollar. Both players simultaneously name shares they would like to have, s_1 and s_2, where $0 \leq s_1, s_2 \leq 1$. If $s_1 + s_2 \leq 1$, then the players receive the shares they named; if $s_1 + s_2 > 1$, then both players receive zero. What are the pure-strategy Nash equilibria of this game?

(a) What is asked. Find all pure-strategy Nash equilibria of the game in which two players simultaneously name shares $s_1, s_2 \in [0, 1]$ of one dollar, and receive them if $s_1 + s_2 \leq 1$ and zero otherwise.

(b) Method. Nash equilibrium with a continuum of strategies (Lecture 2, slide 14; Gibbons §1.1.C, pp. 8–14). The payoffs are discontinuous at $s_1 + s_2 = 1$, so there is no first-order condition: for each class of profiles either show that no deviation pays (equilibrium) or exhibit one that does.

(c) Solution. **Payoffs.** $(u_1, u_2) = (s_1, s_2)$ if $s_1 + s_2 \leq 1$; $(0, 0)$ otherwise.

Step 1: profiles with $s_1 + s_2 = 1$ are equilibria. Player 1 gets s_1. Naming $s_1' < s_1$ gives him only $s_1' < s_1$; naming $s_1' > s_1$ makes the sum exceed 1, so he gets $0 \leq s_1$. So there is no profitable deviation; the same holds for player 2. Every point of the line $\{(s, 1 - s) : 0 \leq s \leq 1\}$ is a Nash equilibrium (the ends $(1, 0)$ and $(0, 1)$ included).

Step 2: profiles with $s_1 + s_2 < 1$ are not equilibria. Player 1 can raise his demand to $s_1' = 1 - s_2 > s_1$; the sum is exactly 1, so he gets $1 - s_2 > s_1$.

Step 3: profiles with $s_1 + s_2 > 1$. Both get 0. If $s_2 < 1$, player 1 deviates to $s_1' = 1 - s_2 > 0$ and gets $1 - s_2 > 0$: not an equilibrium. By symmetry the same holds if $s_1 < 1$. The only profile left is $s_1 = s_2 = 1$: then every deviation by player 1 to $s_1' \in [0, 1)$ still gives sum $s_1' + 1 \geq 1$; for $s_1' > 0$ the sum exceeds 1 and he gets 0; for $s_1' = 0$ the sum is 1 and he gets $s_1' = 0$. No deviation gives more than the current 0. So $(1, 1)$ is a Nash equilibrium, a "weak" one (both payoffs are zero and each player is indifferent among all his strategies).

Answer. The pure-strategy Nash equilibria are all pairs with $s_1 + s_2 = 1$ (a continuum, including $(1, 0)$ and $(0, 1)$) and, in addition, the profile $s_1 = s_2 = 1$ in which both demand the whole dollar and get nothing.

(d) Interpretation. With a continuum of strategies the equilibrium set is a whole line: any split that exhausts the dollar is self-enforcing, because each player prefers to keep his share rather than risk overreaching. Nash equilibrium, as a pure consistency requirement, says nothing about *which* split; a selection device (a focal point such as $\frac{1}{2}$, or bargaining protocol, Lecture 6) is needed. The profile $(1, 1)$ is a coordination failure supported only by indifference.

(e) Verification. `check_L2.py` discretises the strategies to multiples of 1/20 (the deviations used above, $1 - s_j$, lie on the grid) and computes all pure equilibria by brute force: they are exactly the 21 pairs with $s_1 + s_2 = 1$ plus $(1, 1)$, 22 in all. Gibbons gives no answer.

(f) Common mistakes.

- Differentiating. Payoffs jump at $s_1 + s_2 = 1$, so best responses are not smooth.
- Missing $(1, 1)$: it is an equilibrium because the payoff is 0 whatever one demands, not because anybody is happy with it.
- Reporting only $s_1 = s_2 = \frac{1}{2}$ (the symmetric split) as "the" equilibrium.

Item 5: Muñoz-Garcia Ch. 2 Exercise 7

Foundations · difficulty 2/5

Statement (verbatim from the list, item 5)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 2, Exercise 7 "Cournot Game of Quantity Competition", p. 48 [PDF p. 61]. Answer in the source: yes, a worked answer follows the exercise. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider an industry with two firms competing in quantities, i.e., Cournot competition. For simplicity, assume that firms are symmetric in costs, $c > 0$, with no fixed costs and that they face a linear inverse demand $p(Q) = a - bQ$, where $a > c$, $b > 0$, and Q denotes aggregate output. Note that the assumption $a > c$ implies that the highest willingness to pay for the first unit is larger than the marginal cost that firms must incur in order to produce the first unit, thus indicating that a positive production level is profitable in this industry. If firms simultaneously and independently select their output level, q_1 and q_2, find the Nash Equilibrium (NE) of the Cournot game of quantity competition.

(a) What is asked. Find the Nash equilibrium of the Cournot duopoly with inverse demand $p(Q) = a - bQ$, $a > c$, $b > 0$, and common constant marginal cost c.

(b) Method. Cournot's model as a simultaneous game in quantities (Lecture 2, slides 20–25; Tadelis §5.2.3, pp. 87–89). Firm i's profit is concave in its own quantity, so its best response solves the first-order condition; the Nash equilibrium is where the two best responses cross (Lecture 2, slide 23).

(c) Solution. **Game.** Players $1, 2$; strategies $q_i \geq 0$; payoff

$$\pi_i(q_i, q_j) = \big(a - b(q_i + q_j)\big)q_i - c\,q_i.$$

Best response. Take q_j as given.

$$\frac{\partial \pi_i}{\partial q_i} = a - 2bq_i - bq_j - c = 0 \quad \Longrightarrow \quad q_i = \frac{a-c}{2b} - \frac{1}{2}\,q_j.$$

The second derivative is $-2b < 0$, so this is a maximum. The expression is negative when $q_j > (a-c)/b$; since $q_i \geq 0$,

$$BR_i(q_j) = \max\Big\{0,\ \frac{a-c}{2b} - \frac{q_j}{2}\Big\},$$

a line that starts at the monopoly output $(a-c)/(2b)$ when $q_j = 0$, falls with slope $-\frac{1}{2}$ (quantities are strategic substitutes) and hits zero at $q_j = (a-c)/b$. The same holds for firm j.

Crossing. Write $k = (a-c)/(2b)$, so $q_1 = k - q_2/2$ and $q_2 = k - q_1/2$ on the interior branch. Substitute the second into the first:

$$q_1 = k - \tfrac{1}{2}(k - \tfrac{1}{2}q_1) = \tfrac{k}{2} + \tfrac{q_1}{4} \implies \tfrac{3}{4}\,q_1 = \tfrac{k}{2} \implies q_1 = \tfrac{2k}{3} = \frac{a-c}{3b},$$

and then $q_2 = k - k/3 = 2k/3$ as well. Both quantities are positive (since $a > c$), so the interior branch of BR_i is the right one at the solution. Aggregate output, price and profit follow:

$$Q^* = \frac{2(a-c)}{3b}, \qquad p^* = a - bQ^* = \frac{a+2c}{3}, \qquad \pi^* = (p^* - c)\,q^* = \frac{a-c}{3} \cdot \frac{a-c}{3b} = \frac{(a-c)^2}{9b}.$$

Uniqueness. The interior equations above are linear with a single solution. A profile with $q_1 = 0$ would need $q_2 = BR_2(0) = k$ and then $BR_1(k) = k/2 > 0 = q_1$, a contradiction; the same for $q_2 = 0$. Hence (q^*, q^*) is the unique Nash equilibrium.

Answer. $q_1^* = q_2^* = \dfrac{a-c}{3b}$, with $Q^* = \dfrac{2(a-c)}{3b}$, $p^* = \dfrac{a+2c}{3}$ and profit $\dfrac{(a-c)^2}{9b}$ for each firm. The book's answer is the same.

(d) Interpretation. Compare with the two benchmarks of the slides (Lecture 2, slide 22). A monopolist produces $(a-c)/(2b)$ and earns $(a-c)^2/(4b)$; perfect competition has $p = c$, $Q = (a-c)/b$. The duopoly sits in between: $Q^* = \frac{2}{3}(a-c)/b$ is more than the monopoly output $\frac{1}{2}(a-c)/b$ and less than the competitive output, and industry profit $2(a-c)^2/(9b)$ is below the monopoly's $(a-c)^2/(4b)$ ($0.222 < 0.25$ in units of $(a-c)^2/b$). Each firm ignores the harm its output does to the other's price, so total output is too high from the cartel's point of view (Lecture 2, slides 25–26).

(e) Verification. `check_L2.py` differentiates symbolically, solves the two first-order conditions with sympy and compares with the book's printed answer $q^* = (a-c)/(3b)$ (and its best-response function $(a-c)/(2b) - q_j/2$). It also recomputes the case $a = 10$, $b = 1$, $c = 2$ (so $q^* = 8/3$, profit $64/9$) by exact search over a grid of quantities, and confirms that the best response is 0 at $q_j = (a-c)/b$.

(f) Common mistakes.

- Maximising *total* profit $\pi_1 + \pi_2$ instead of each firm's own profit: that is the cartel, $q_i = (a-c)/(4b)$.
- Forgetting that q_j also enters the price, so the coefficient on q_i in the first-order condition is $2b$, not b.
- Dropping the corner: the true best response is $\max\{0, \cdot\}$.

Item 6: Muñoz-Garcia Ch. 3, Exercise 1

Foundations · difficulty 2/5

Statement (verbatim from the list, item 6)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 3, Exercise 1 "Game of Chicken", pp. 86–87 [PDF p. 99–100]. Answer in the source: yes, a worked answer follows the exercise. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider the game of Chicken (depicted in Fig. 3.1), in which two players driving their cars against each other must decide whether or not to swerve.
>
> **Part (a)** Is there any strictly dominated strategy for Player 1? And for Player 2?
>
> **Part (b)** What are the best responses for Player 1? And for Player 2?
>
> **Part (c)** Can you find any pure strategy Nash Equilibrium (psNE) in this game?
>
> **Part (d)** Find the mixed strategy Nash Equilibrium (msNE) of the game. *Hint*: denote by p the probability that Player 1 chooses Straight and by $(1-p)$ the probability that he chooses to Swerve. Similarly, let q denote the probability that Player 2 chooses Straight and $(1-q)$ the probability that she chooses to Swerve.
>
> **Part (e)** Show your result of part (d) by graphically representing every player i's best response function $BRF_i(s_j)$, where $s_j = \{Swerve, Straight\}$ is the strategy selected by player $j \neq i$.

Player 1 \ *Player 2*	Straight	Swerve
Straight	0,0	3,1
Swerve	1,3	2,2

Fig. 3.1 Game of Chicken

(a) What is asked. For Chicken, find dominated strategies, best responses, the pure and the mixed Nash equilibria, and draw the best-response functions.

(b) Method. Dominance (Lecture 1, slide 23; Tadelis §4.1, pp. 59–61); best responses and pure NE (Tadelis §4.3.1, p. 69, and §5.1, p. 80; Lecture 2, slides 10 and 14); mixed strategies and expected payoffs (Tadelis §6.1, p. 105; Lecture 2, slides 42–43); mixed NE through the principle of indifference (Tadelis §6.2 and Prop. 6.1, pp. 107–108; Lecture 2, slides 44–46): a player mixes only if the other's mix makes him indifferent. The graph in (e) follows the battle-of-the-sexes example (Lecture 2, slides 47–50).

(c) Solution. Notation. St = Straight, Sw = Swerve. Matrix (Fig. 3.1): $(St, St) = (0, 0)$, $(St, Sw) = (3, 1)$, $(Sw, St) = (1, 3)$, $(Sw, Sw) = (2, 2)$. The game is symmetric.

(a) Player 1's rows over columns (St, Sw): $St = (0, 3)$, $Sw = (1, 2)$. Sw is better against St $(1 > 0)$, St is better against Sw $(3 > 2)$: neither row dominates the other. By symmetry the same holds for player 2. No strictly dominated strategy.

(b) Best responses underlined:

$1 \setminus 2$	St	Sw
St	0, 0	$\underline{3}$, $\underline{1}$
Sw	$\underline{1}$, $\underline{3}$	2, 2

$BR_1(St) = Sw$ (payoff $1 > 0$) and $BR_1(Sw) = St$ (payoff $3 > 2$). By symmetry $BR_2(St) = Sw$ and $BR_2(Sw) = St$. Each player wants the opposite of the other: an anti-coordination game.

(c) A cell is a pure NE when both strategies are best responses to each other. (St, Sw): $St = BR_1(Sw)$ and $Sw = BR_2(St)$, NE. (Sw, St): NE by symmetry. (St, St) and (Sw, Sw) are not, since in each both players would switch. The psNE are (St, Sw) with payoffs $(3, 1)$ and (Sw, St) with payoffs $(1, 3)$.

(d) Mixed equilibrium. Player 1's expected payoffs when player 2 plays St with probability q:

$$EU_1(St) = 0 \cdot q + 3(1 - q) = 3 - 3q, \tag{1}$$
$$EU_1(Sw) = 1 \cdot q + 2(1 - q) = 2 - q. \tag{2}$$

Player 1 is willing to mix only if (1) = (2):

$$3 - 3q = 2 - q \implies 1 = 2q \implies q^* = \tfrac{1}{2}.$$

Player 2's expected payoffs when player 1 plays St with probability p (her payoffs are 0 at (St, St), 3 at (Sw, St), 1 at (St, Sw), 2 at (Sw, Sw)):

$$EU_2(St) = 0 \cdot p + 3(1 - p) = 3 - 3p, \tag{3}$$
$$EU_2(Sw) = 1 \cdot p + 2(1 - p) = 2 - p. \tag{4}$$

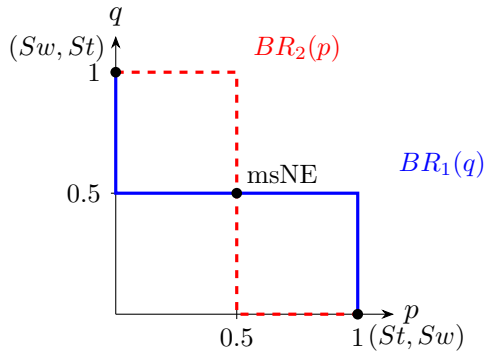
Setting (3) = (4): $3 - 3p = 2 - p$, so $p^* = \frac{1}{2}$. The msNE is

$$\sigma^* = \Big((\tfrac{1}{2}\, St, \tfrac{1}{2}\, Sw),\ (\tfrac{1}{2}\, St, \tfrac{1}{2}\, Sw) \Big),$$

and each player's expected payoff is $EU_1 = 3 - 3 \cdot \frac{1}{2} = 1.5$ (by (1) at $q^* = \frac{1}{2}$).

(e) Best-response functions. From (1)–(2), $EU_1(St) - EU_1(Sw) = 1 - 2q$, so

$$BR_1(q) = \begin{cases} p = 1 & q < \frac{1}{2}, \\ p \in [0, 1] & q = \frac{1}{2}, \\ p = 0 & q > \frac{1}{2}, \end{cases} \qquad BR_2(p) = \begin{cases} q = 1 & p < \frac{1}{2}, \\ q \in [0, 1] & p = \frac{1}{2}, \\ q = 0 & p > \frac{1}{2}. \end{cases}$$



The two graphs cross three times: at $(p, q) = (1, 0)$, i.e. (St, Sw); at $(0, 1)$, i.e. (Sw, St); and at $(\frac{1}{2}, \frac{1}{2})$, the msNE.

(d) Interpretation. Chicken has the shape of Hawk-Dove (Tadelis 5.6). In the mixed equilibrium each driver goes straight half of the time, so they crash with probability $\frac{1}{4}$, and each expects 1.5, less than the 2 of mutual swerving and less than the 3 of the favourable pure equilibrium. The mix is pinned by the *other* player's indifference, not by one's own payoffs (Lecture 2, slide 50). Raising one's own payoff from swerving would not change one's own mix; it would change the *opponent's*.

(e) Verification. `check_A2.py` checks (a)–(c) by brute force, solves both indifference conditions with sympy ($q^* = p^* = \frac{1}{2}$, payoff $\frac{3}{2}$ for both players, crash probability $\frac{1}{4}$), and finds on a grid of $(p, q) \in \{0, \frac{1}{100}, \ldots, 1\}^2$ that the mutual best responses are exactly $(1, 0)$, $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$. *Comparison with the book's answer* (pp. 87–91): it agrees on all five parts (no dominated strategies, the same best responses, the two psNE, msNE with $p = q = \frac{1}{2}$, the same best-response graph, Figs. 3.4–3.6). One typo in the book: on p. 89 the rearranged indifference condition is printed "$3+3q = q+2-2q \Rightarrow q = 1/2$"; from $0q+3(1-q) = 1q+2(1-q)$ it should read $3-3q = q+2-2q$ (as printed, $3+3q = 2-q$ gives $q = -\frac{1}{4}$). The book's result $q = \frac{1}{2}$ is correct.

(f) Common mistakes.

- Solving player 1's indifference for p. Player 1's indifference condition contains only q: it pins down the *opponent's* mix.
- Copying the book's typo line $3+3q = q+2-2q$; it gives $q = -\frac{1}{4}$, which is not a probability and should be a warning sign.
- Drawing BR_1 as a function of p on the horizontal axis. BR_1 maps q to p; with p on the horizontal axis its graph has a horizontal segment at $q = \frac{1}{2}$.

Answer. (a) No strictly dominated strategy for either player. (b) $BR_i(St) = Sw$, $BR_i(Sw) = St$. (c) psNE: (St, Sw) and (Sw, St). (d) msNE: each plays Straight with probability $\frac{1}{2}$ ($p^* = q^* = \frac{1}{2}$), expected payoff 1.5 each. (e) $BR_1(q)$: $p = 1$ for $q < \frac{1}{2}$, $p = 0$ for $q > \frac{1}{2}$, any p at $q = \frac{1}{2}$ (symmetric for player 2); the graphs cross at $(1, 0)$, $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$.

Item 7: Tadelis Exercise 6.1

Foundations · difficulty 2/5

Statement (verbatim from the list, item 7)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 6.1 "Best Responses in the Battle of the Sexes", p. 123 [PDF p. 142]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim.

> **Best Responses in the Battle of the Sexes:** Use the best-response correspondences in the Battle of the Sexes game to find all the Nash equilibria. (Follow the approach used for the example in Section 6.2.3.)

Referred to in the statement. The Battle of the Sexes (Tadelis §3.3, pp. 53–54 [PDF p. 72–73]) (reproduced from the book):

> Alex and Chris are a couple, and they need to choose where to meet this evening, with no means of communicating. Both prefer being together over not being together, but Alex prefers opera (O) to football (F), while Chris prefers the opposite.

Alex \ Chris	O	F
O	2, 1	0, 0
F	0, 0	1, 2

(a) What is asked. Use the best-response correspondences to find all Nash equilibria, pure and mixed, of the Battle of the Sexes (O = opera, F = football; Alex chooses the row, Chris the column; payoffs $(2,1)$ at (O,O), $(1,2)$ at (F,F), $(0,0)$ when they miss each other).

(b) Method. Mixed strategies and the principle of indifference (Lecture 2, slides 42–46), then the best-response graph (Lecture 2, slides 47–50, "Example 3"; Tadelis §6.2.3, pp. 113–114). A mixed equilibrium is a point where the two best-response correspondences cross.

(c) Solution. Let $p = \Pr(\text{Alex plays } O)$ and $q = \Pr(\text{Chris plays } O)$.

Alex \ Chris	O	F
O	2, 1	0, 0
F	0, 0	1, 2

Alex's best response. His expected payoffs against Chris's q are

$$v_A(O,q) = 2q, \qquad v_A(F,q) = 1-q, \qquad v_A(O,q) - v_A(F,q) = 3q - 1.$$

So O is strictly better iff $q > \frac{1}{3}$, F iff $q < \frac{1}{3}$, and he is indifferent at $q = \frac{1}{3}$:

$$BR_A(q) = \begin{cases} p = 0 & q < \frac{1}{3}, \\ p \in [0,1] & q = \frac{1}{3}, \\ p = 1 & q > \frac{1}{3}. \end{cases}$$

Chris's best response. Against Alex's p:

$$v_C(p,O) = p, \qquad v_C(p,F) = 2(1-p), \qquad v_C(p,O) - v_C(p,F) = 3p - 2,$$

so

$$BR_C(p) = \begin{cases} q = 0 & p < \frac{2}{3}, \\ q \in [0,1] & p = \frac{2}{3}, \\ q = 1 & p > \frac{2}{3}. \end{cases}$$

Crossing the two graphs in the unit square (p, q) (Alex's is a step at $q = \frac{1}{3}$, Chris's a step at $p = \frac{2}{3}$):

- $q > \frac{1}{3}$ forces $p = 1 > \frac{2}{3}$, which forces $q = 1$: the profile $(p, q) = (1, 1)$, that is (O, O).
- $q < \frac{1}{3}$ forces $p = 0 < \frac{2}{3}$, which forces $q = 0$: $(0, 0)$, that is (F, F).
- $q = \frac{1}{3}$ lets p be anything, but then Chris needs $p = \frac{2}{3}$ to be willing to mix: $(p, q) = (\frac{2}{3}, \frac{1}{3})$.

These are the only three crossing points: the two pure equilibria and one mixed equilibrium in which Alex plays O with probability $\frac{2}{3}$ and Chris plays O with probability $\frac{1}{3}$ (F with probability $\frac{2}{3}$).

Payoffs at the mixed equilibrium. Alex: $v_A = 2q = \frac{2}{3}$; Chris: $v_C = p = \frac{2}{3}$. They meet with probability $pq + (1-p)(1-q) = \frac{2}{9} + \frac{2}{9} = \frac{4}{9}$, and each earns only $\frac{2}{3}$, below the worst pure equilibrium payoff 1.

Answer. Three Nash equilibria: (O, O), (F, F), and the mixed one $(\sigma_A, \sigma_C) = ((\frac{2}{3}O, \frac{1}{3}F),\ (\frac{1}{3}O, \frac{2}{3}F))$, with expected payoff $\frac{2}{3}$ for each player.

(d) Interpretation. Each player's mixing probability makes the *other* indifferent, not himself: Alex plays his favourite O with probability $\frac{2}{3}$ because that is exactly what keeps Chris, who favours F, indifferent. The mixed equilibrium is worse for both than either pure one, and it is the third (odd) equilibrium that "almost always" accompanies two pure ones in a 2×2 game (Tadelis §6.2.3, footnote 6; Nash's theorem, Lecture 2, slide 51).

(e) Verification. `check_L2.py` solves the two indifference conditions with sympy, then enumerates mutual best responses over a grid of 61×61 exact fractions: the only crossings are $(0, 0)$, $(\frac{2}{3}, \frac{1}{3})$ and $(1, 1)$. It also recomputes the payoffs $\frac{2}{3}$ and the meeting probability $\frac{4}{9}$. Tadelis gives no answer.

(f) Common mistakes.

- Equalising one's *own* payoffs: Alex's $p = \frac{2}{3}$ comes from Chris's indifference (equation $p = 2(1-p)$), not from Alex's.
- Reading $q = \frac{1}{3}$ as the probability that Chris plays F. Here q is the probability of O throughout.
- Forgetting the two pure equilibria, which are the end points of the same graphs.

Item 8: Gibbons Problem 1.12

Exam · difficulty 2/5

Statement (verbatim from the list, item 8)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §1.3 Problems, Problem 1.12, p. 51 [PDF p. 59]. Answer in the source: none (Gibbons prints no answers). Reproduced verbatim.

1.12. Find the mixed-strategy Nash equilibrium of the following normal-form game.

	L	R
T	2,1	0,2
B	1,2	3,0

(a) What is asked. Find the mixed-strategy Nash equilibrium of the 2×2 game with T, B against L, R.

(b) Method. Mixed strategies and expected payoffs (Gibbons §1.3.A, pp. 29–33; Lecture 2, slides 42–43); mixed Nash equilibrium through the principle of indifference (Gibbons §1.3.B, pp. 33–44; Tadelis Prop. 6.1, p. 108; Lecture 2, slides 44–46). First check that there is no pure equilibrium, so any equilibrium must have both players mixing.

(c) Solution. **Best responses**, underlined:

$1 \setminus 2$	L	R
T	$\underline{2}$, 1	0, $\underline{2}$
B	1, $\underline{2}$	$\underline{3}$, 0

$BR_1(L) = \{T\}$ $(2 > 1)$, $BR_1(R) = \{B\}$ $(3 > 0)$; $BR_2(T) = \{R\}$ $(2 > 1)$, $BR_2(B) = \{L\}$ $(2 > 0)$. No cell has both payoffs underlined: the best responses cycle $(T, L) \to (T, R) \to (B, R) \to (B, L) \to (T, L)$, so there is no pure-strategy Nash equilibrium.

No equilibrium with only one player mixing. If player 2 played a pure strategy, player 1's best reply would be unique and pure, and then player 2's best reply to it would be the other column; so in equilibrium player 2 mixes, and by the same argument so does player 1. Both must be indifferent between their two pure strategies (Lecture 2, slide 44).

Player 2's mix, from player 1's indifference. Let $q = \Pr(L)$.

$$EU_1(T) = 2q + 0(1-q) = 2q, \qquad EU_1(B) = 1 \cdot q + 3(1-q) = 3 - 2q.$$

$$2q = 3 - 2q \implies 4q = 3 \implies q^* = \tfrac{3}{4}.$$

Player 1's mix, from player 2's indifference. Let $p = \Pr(T)$.

$$EU_2(L) = 1 \cdot p + 2(1-p) = 2 - p, \qquad EU_2(R) = 2p + 0(1-p) = 2p.$$

$$2 - p = 2p \implies 3p = 2 \implies p^* = \tfrac{2}{3}.$$

Equilibrium and payoffs.

$$\sigma^* = \Big(\big(\tfrac{2}{3}\,T,\ \tfrac{1}{3}\,B\big),\ \big(\tfrac{3}{4}\,L,\ \tfrac{1}{4}\,R\big)\Big), \qquad u_1^* = 2q^* = \tfrac{3}{2}, \qquad u_2^* = 2p^* = \tfrac{4}{3}.$$

Uniqueness. $EU_1(T) - EU_1(B) = 4q - 3$, so $BR_1(q)$ is $p = 1$ for $q > \frac{3}{4}$, $p = 0$ for $q < \frac{3}{4}$, and $[0, 1]$ at $q = \frac{3}{4}$. $EU_2(L) - EU_2(R) = 2 - 3p$, so $BR_2(p)$ is $q = 1$ for $p < \frac{2}{3}$, $q = 0$ for $p > \frac{2}{3}$, and $[0, 1]$ at $p = \frac{2}{3}$. The graphs cross only at $(p, q) = (\frac{2}{3}, \frac{3}{4})$: it is the unique Nash equilibrium.

(d) Interpretation. This is a Matching-Pennies-like game (Gibbons §1.3.A): player 1 wants to match (T with L, B with R), player 2 wants to mismatch. Each player randomises so as to leave the *other* indifferent, which is why player 1's probabilities come from player 2's payoffs. For instance, raising player 1's payoff at (B, R) from 3 to 4 would leave p^* unchanged and raise q^* to $\frac{4}{5}$ (from $2q = 4 - 3q$).

(e) Verification. `check_A2.py` confirms by brute force that there is no pure equilibrium, solves both indifference conditions with sympy ($q^* = \frac{3}{4}$, $p^* = \frac{2}{3}$), checks the payoffs $\frac{3}{2}$ and $\frac{4}{3}$ and the comparative-statics remark, and finds on a grid of mixed profiles that $(\frac{2}{3}, \frac{3}{4})$ is the only mutual best response. Gibbons gives no answer.

(f) Common mistakes.

- Solving player 1's indifference for p. The condition $EU_1(T) = EU_1(B)$ contains only q, player 2's probability.
- Reporting $(\frac{3}{4}, \frac{2}{3})$ in the wrong order. State which probability belongs to which player and which strategy.
- Stopping at "no pure NE". Nash's theorem (Lecture 2, slide 51) guarantees a mixed one, and the question asks for it.

Answer. No pure-strategy equilibrium. The unique (mixed) Nash equilibrium: player 1 plays T with probability $\frac{2}{3}$ and B with $\frac{1}{3}$; player 2 plays L with probability $\frac{3}{4}$ and R with $\frac{1}{4}$. Expected payoffs $\frac{3}{2}$ for player 1 and $\frac{4}{3}$ for player 2.

Item 9: Tadelis Exercise 5.1

Exam · difficulty 3/5

Statement (verbatim from the list, item 9)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 5.1, p. 95 [PDF p. 114]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim.

> **Prove Proposition 5.1.**

Referred to in the statement. Proposition 5.1 (Tadelis pp. 80–81 [PDF p. 99–100]) (reproduced from the book):

> **Proposition 5.1** Consider a strategy profile $s = (s_1, s_2, \ldots, s_n)$. If s is either
>
> 1. a strict dominant strategy equilibrium,
> 2. the unique survivor of IESDS, or
> 3. the unique rationalizable strategy profile,
>
> then s is the unique Nash equilibrium.

(a) What is asked. Prove Proposition 5.1: if a strategy profile s is a strict dominant-strategy equilibrium, or the unique survivor of IESDS, or the unique rationalizable profile, then s is the unique Nash equilibrium.

(b) Method. Direct arguments. For (1) and (2), a profitable deviation against s_{-i} forces the deviating strategy to have been deleted at some round; for (3), the definition of rationalizability applied to a singleton. The slides prove (2) in two halves, Propositions 1 and 2 (Lecture 2, slides 16–17), and the proof below follows the same "first round in which something is deleted" argument. All games are finite; "Nash equilibrium" means a pure-strategy one, as in Tadelis §5.1.

(c) Solution. Part 1: strict dominant-strategy equilibrium. Let s_i^* be strictly dominant for every i: $v_i(s_i^*, s_{-i}) > v_i(s_i, s_{-i})$ for all $s_i \neq s_i^*$ and all s_{-i}.

- *s^* is a Nash equilibrium.* Take $s_{-i} = s_{-i}^*$: the inequality says s_i^* does strictly better than any other s_i, so it is a best reply to s_{-i}^* for every i.
- *It is the only one.* If $s \neq s^*$ were a Nash equilibrium, some $s_i \neq s_i^*$ would be a best reply to s_{-i}, contradicting $v_i(s_i^*, s_{-i}) > v_i(s_i, s_{-i})$.

(Part 1 also follows from Part 2: a strict dominant strategy deletes all the others in round 1.)

Part 2: unique survivor of IESDS. Let S^k be the strategy sets after k rounds and let s^* be the only profile in S^∞.

- *s^* is a Nash equilibrium (Proposition 2).* Suppose player i had a strategy s_i with $v_i(s_i, s_{-i}^*) > v_i(s_i^*, s_{-i}^*)$. Then $s_i \neq s_i^*$, so s_i was deleted in some round k: there is $s_i' \in S_i^{k-1}$ with $v_i(s_i', t_{-i}) > v_i(s_i, t_{-i})$ for all $t_{-i} \in S_{-i}^{k-1}$. The profile s_{-i}^* lies in S_{-i}^{k-1} (it survives forever), so

$$v_i(s_i', s_{-i}^*) > v_i(s_i, s_{-i}^*) > v_i(s_i^*, s_{-i}^*),$$

and $s_i' \neq s_i^*$ is again a strictly better reply to s_{-i}^*. Apply the same argument to s_i', and so on. The values $v_i(\cdot, s_{-i}^*)$ rise strictly along the chain, so no strategy repeats, and the strategy sets are finite: the chain must stop at a strategy that is never deleted, that is, at s_i^*. But every member of the chain does strictly better against s_{-i}^* than s_i^* does, a contradiction. So no profitable deviation exists.

- *It is the only Nash equilibrium (Proposition 1).* Let s be a Nash equilibrium and suppose it does not survive. Take the first round k in which some component s_i is deleted: s_i is strictly dominated by some $s_i' \in S_i^{k-1}$ against every $t_{-i} \in S_{-i}^{k-1}$, and $s_{-i} \in S_{-i}^{k-1}$ because no component was deleted earlier. Then $v_i(s_i', s_{-i}) > v_i(s_i, s_{-i})$, so s_i is not a best reply to s_{-i}, a contradiction. Hence every Nash equilibrium lies in $S^\infty = \{s^*\}$.

Part 3: unique rationalizable profile. Let R^k be the strategy sets after k rounds of deleting strategies that are never a best reply to a belief over the opponents' surviving strategies, and let $R^\infty = \{s^*\}$.

- s^* *is a Nash equilibrium.* The sets are finite and decrease to R^∞, so for k large $R_{-i}^{k-1} = \{s_{-i}^*\}$. Since s_i^* survives round k, it is a best reply to some belief over R_{-i}^{k-1}, and the only such belief is the point mass on s_{-i}^*. So s_i^* is a best reply to s_{-i}^*, for each i.
- *It is the only one: every Nash equilibrium is rationalizable.* Let s be a Nash equilibrium. By induction on k, $s \in R^k$: it holds for $k = 0$; if $s_{-i} \in R_{-i}^{k-1}$, then s_i is a best reply to the belief "s_{-i} with probability one", which is a belief over R_{-i}^{k-1}, so $s_i \in R_i^k$. Hence $s \in R^\infty = \{s^*\}$.

■

(d) Interpretation. The three hypotheses are three ways of saying that rational, mutually aware players have a unique prediction, and they are nested: a strict dominant-strategy equilibrium is the unique survivor of IESDS (everything else goes in round 1), and a unique IESDS survivor is the unique rationalizable profile (MWG exercise 8.C.1). So Nash equilibrium is a *necessary* condition for any of them to make a single prediction; this is why the slides treat IESDS and rationalizability as the epistemic benchmark and Nash equilibrium as the sharper concept (Lecture 2, slides 15–17). The converse fails: the Battle of the Sexes has two Nash equilibria and IESDS deletes nothing.

(e) Verification. `check_L2.py` tests the proposition on random finite two-player games (1,200 games with 2–4 strategies per player and payoffs in $0, \ldots, 5$, ties included): whenever the game has a strict dominant-strategy equilibrium, a unique IESDS survivor (pure dominators), or a unique rationalizable profile (computed by linear programming over mixed beliefs), the set of pure Nash equilibria is exactly that profile. It also confirms the lemma that every pure Nash equilibrium is rationalizable. The proof itself is the argument above. Tadelis gives no answer.

(f) Common mistakes.

- Proving only "s is a Nash equilibrium" and forgetting uniqueness, which is the other half (Proposition 1).
- Using *weak* dominance: a weakly dominant-strategy equilibrium can coexist with other Nash equilibria (exercise 5.2).
- Assuming that the strategy deleted in some round is dominated by s_i^* itself; the dominator may be deleted later, which is why the chain argument is needed.

Item 10: Tadelis Exercise 5.2

Exam · difficulty 2/5

Statement (verbatim from the list, item 10)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 5.2 "Weak Dominance", p. 95 [PDF p. 114]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim.

> **Weak Dominance:** A strategy $s^W \in S$ is a weakly dominant strategy equilibrium if $s_i^W \in S_i$ is a weakly dominant strategy for all $i \in N$, that is, if $v_i(s_i^W, s_{-i}) \geq v_i(s_i', s_{-i})$ for all $s_i' \in S_i$ and for all $s_{-i} \in S_{-i}$. Provide an example of a game for which there is a weakly dominant strategy equilibrium as well as another Nash equilibrium.

(a) What is asked. Give an example of a game with a weakly dominant-strategy equilibrium and another Nash equilibrium.

(b) Method. Construct and verify. Weak dominance only requires "at least as good against everything", so ties against some opponent strategies are allowed, and the ties are where a second equilibrium can hide (Lecture 2, slide 16: "a Nash equilibrium may involve weakly dominated strategies"; Tadelis §5.1, p. 81). Take the smallest game with a tie: 2×2, with a coordination payoff in one cell only.

(c) Solution. **The game.** Player 1 chooses U or D, player 2 chooses L or R.

$1 \setminus 2$	L	R
U	$\underline{1}, \underline{1}$	$\underline{0}, 0$
D	$0, \underline{0}$	$\underline{0}, \underline{0}$

(Underlined payoffs: the payoff is a best reply of its player to the other's strategy.)

U and L are weakly dominant. For player 1: against L, U gives $1 > 0$; against R, U gives $0 = 0$. So $v_1(U, s_2) \geq v_1(D, s_2)$ for both s_2, with a strict inequality against L. For player 2: against U, L gives $1 > 0$; against D, L gives $0 = 0$. So (U, L) is a weakly dominant-strategy equilibrium, and it is a Nash equilibrium (the definition of weak dominance applied at $s_{-i} = s_{-i}^W$).

A second Nash equilibrium: (D, R). Against R, player 1 gets 0 from D and 0 from U: no gain from deviating. Against D, player 2 gets 0 from R and 0 from L: no gain either. So neither player can strictly gain by deviating, and (D, R) is a Nash equilibrium.

No other profile is an equilibrium: at (U, R) player 2 gains by switching to L ($1 > 0$) and at (D, L) player 1 gains by switching to U.

Answer. In the game above, (U, L) is a weakly dominant-strategy equilibrium, and (D, R), in which both players use weakly dominated strategies, is another Nash equilibrium. The two equilibria give payoffs $(1, 1)$ and $(0, 0)$.

(d) Interpretation. Weak dominance does not pin down a unique equilibrium, whereas strict dominance does (exercise 5.1, Part 1). At (D, R) each player is indifferent between his two actions,

so deviating costs nothing, but it does not help either: the equilibrium survives only on ties. This is why the slides call such equilibria "less plausible" (Lecture 2, slide 16) and why refinements (trembling hands, eliminating weakly dominated strategies) discard them. The equilibrium (U, L) is the one a player who is unsure about the other's move would pick: U and L are never worse.

(e) Verification. `check_L2.py` enumerates the four cells: the pure Nash equilibria are exactly (U, L) and (D, R); U and L weakly dominate ($\geq$ everywhere, $>$ somewhere); and the payoffs at (D, R) are ties, not strict losses. Tadelis gives no answer.

(f) Common mistakes.

- Using strict inequalities everywhere: with strict dominance the equilibrium would be unique.
- Giving a game in which the second equilibrium is not an equilibrium after all, because a deviation is strictly profitable; check every cell.
- Forgetting that the definition applies to *each* player: both s_1^W and s_2^W must be weakly dominant.

Item 11: Tadelis Exercise 5.3

Exam · difficulty 4/5

Statement (verbatim from the list, item 11)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 5.3 "Nash and IESDS", pp. 95–96 [PDF p. 114–115]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim.

> **Nash and IESDS:** Consider a two-player game with m pure strategies for each player that can be represented by an $m \times m$ matrix.
>
> a. Show that if $m = 2$ and the game has a unique pure-strategy Nash equilibrium then this is the unique strategy profile that survives IESDS.
>
> b. Show that if $m = 3$ and the game has a unique pure-strategy equilibrium then it may not be the only strategy profile that survives IESDS.

(a) What is asked. (a) Show that in an $m \times m$ game with $m = 2$ and a unique pure-strategy Nash equilibrium, that equilibrium is the unique IESDS survivor. (b) Show that for $m = 3$ it need not be.

(b) Method. The relation between Nash equilibrium and IESDS (Lecture 2, slides 15–17). By Proposition 1 (slide 16) every Nash equilibrium survives IESDS, so the question is whether *more* than the equilibrium survives. In a 2×2 game a strategy is strictly dominated iff the other one is a strictly better reply to *both* opponent strategies, so everything can be read off the best replies $b_1(L), b_1(R)$ of player 1 and $b_2(U), b_2(D)$ of player 2.

A caveat on part (a). As literally stated (payoffs with possible ties) part (a) is not true. The 2×2 game with payoffs

$1 \setminus 2$	L	R
U	$\underline{1},\ \underline{1}$	$\underline{1},\ 0$
D	$\underline{1},\ 0$	$0,\ \underline{1}$

has the unique pure-strategy Nash equilibrium (U, L): at (U, R) player 2 earns 0 and L gives 1; at (D, L) player 2 earns 0 and R gives 1; at (D, R) player 1 earns 0 and U gives 1. Yet neither U nor D is strictly dominated (they tie against L), and neither L nor R is dominated (L is better against U, R against D), so IESDS deletes nothing and all four profiles survive. The proof below therefore assumes the generic case with no ties between a player's payoffs from his two strategies against the same opponent strategy, which is what the exercise intends (it is also the case in which best replies are unique); the counterexample shows the assumption is needed.

(c) Solution. (a) The proof, with no ties. Write the rows U, D and the columns L, R. With no ties, each player has a unique best reply to each strategy of the opponent.

- *Case 1:* $b_1(L) = b_1(R) = \hat{r}$. Then $\hat{r}$ is strictly better than the other row against both columns, so the other row is deleted in round 1. Player 2 now faces one row, so one of his columns, $\hat{c} = b_2(\hat{r})$, is strictly better than the other and the other is deleted in round 2. IESDS ends at the single profile $(\hat{r}, \hat{c})$, which is a Nash equilibrium ($\hat{r}$ is a best reply to $\hat{c}$ and $\hat{c}$ to $\hat{r}$). By uniqueness it is the given equilibrium.

- *Case 2:* $b_2(U) = b_2(D)$. Symmetric: a column is strictly dominant, then a row.
- *Case 3:* $b_1(L) \neq b_1(R)$ *and* $b_2(U) \neq b_2(D)$. Then neither player has a strictly dominated strategy and the game is not reduced. But b_1 and b_2 are bijections between $\{U, D\}$ and $\{L, R\}$, so the composition $\varphi = b_2 \circ b_1$ is a permutation of $\{L, R\}$. A profile $(b_1(c), c)$ is a Nash equilibrium iff $b_2(b_1(c)) = c$, i.e. iff c is a fixed point of φ. A permutation of two elements is the identity (two fixed points: two equilibria) or the swap (none). So the game has 0 or 2 pure equilibria and cannot have exactly one.

Hence, when the equilibrium is unique, only Cases 1–2 occur, and IESDS leaves exactly the equilibrium. ■

(b) A 3×3 game. Rows a, b, c; columns L, M, R.

$1 \setminus 2$	L	M	R
a	$\underline{2}, \underline{2}$	$0, 0$	$0, 0$
b	$0, 0$	$\underline{2}, 0$	$0, \underline{2}$
c	$0, 0$	$0, \underline{2}$	$\underline{2}, 0$

Best replies. Player 1: $b_1(L) = a$, $b_1(M) = b$, $b_1(R) = c$. Player 2: $b_2(a) = L$, $b_2(b) = R$, $b_2(c) = M$. Every best reply is unique (payoff 2 against 0).

Unique pure Nash equilibrium. A profile is an equilibrium iff it is underlined for both players: only (a, L) has both payoffs underlined. For the other eight cells check one deviation: at (b, M) player 2 prefers R; at (b, R) player 1 prefers c; at (c, M) player 1 prefers b; at (c, R) player 2 prefers M; and in the first row and column, (a, M) and (a, R) have player 2 preferring L, and (b, L) and (c, L) have player 1 preferring a.

IESDS deletes nothing. Every strategy is the unique best reply to some pure strategy of the opponent (a to L, b to M, c to R, and L to a, R to b, M to c). A strategy that is a best reply to some belief cannot be strictly dominated, by a pure or a mixed strategy. So all nine profiles survive, although only (a, L) is a Nash equilibrium.

Answer. (a) With no ties, a 2×2 game with exactly one pure Nash equilibrium is dominance-solvable (Cases 1–2) or has 0 or 2 pure equilibria (Case 3), so the equilibrium is the unique IESDS survivor; with ties the claim can fail (the counterexample above). (b) In the 3×3 game above, (a, L) is the unique pure Nash equilibrium, yet no strategy is strictly dominated, so all nine profiles survive IESDS.

(d) Interpretation. Nash equilibrium refines IESDS (Proposition 1), and in 2×2 games the refinement is all-or-nothing: either dominance solves the game, or the best replies form a "cycle" with an even number of fixed points. With three strategies a best-reply *cycle* (the sub-game in $\{b, c\} \times \{M, R\}$ is matching pennies) can sit beside an equilibrium, and IESDS cannot touch it. The game of (b) also has a mixed equilibrium in that cycle, so the unique *pure* equilibrium is not the unique equilibrium; the exercise only asks about pure ones.

(e) Verification. `check_L2.py` enumerates all $4^8 = 65{,}536$ ordinal 2×2 games (payoffs in $\{0, 1, 2, 3\}$): every pure equilibrium survives IESDS; among games with no ties of the kind above, no game with exactly one equilibrium keeps more than one profile, and Case 3 games always have 0 or 2 equilibria; among games with ties there are games that violate (a) (the counterexample is one). For (b) it checks the equilibrium set, the best-reply bijections, and, by linear programming, that no strategy is strictly dominated even by a mixture. Tadelis gives no answer.

(f) Common mistakes.

- Arguing (a) only through the Case 1 pattern (dominance-solvable) and forgetting Case 3.
- In (b), producing a game with several equilibria: the exercise wants *one* pure equilibrium and extra survivors.
- Using weak dominance to prune in (b); IESDS is strict.

Item 12: Tadelis Exercise 5.8

Exam · difficulty 3/5

Statement (verbatim from the list, item 12)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 5.8 "The n-Firm Cournot Model", pp. 96–97 [PDF p. 115–116]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim (the printed "$Q = q_i + \cdots + q_n$" is kept as printed).

> **The n-Firm Cournot Model:** Suppose there are n firms in the Cournot oligopoly model. Let q_i denote the quantity produced by firm i, and let $Q = q_i + \cdots + q_n$ denote the aggregate production. Let $P(Q)$ denote the market clearing price (when demand equals Q) and assume that the inverse demand function is given by $P(Q) = a - Q$, where $Q \leq a$. Assume that firms have no fixed cost and that the cost of producing quantity q_i is cq_i (all firms have the same marginal cost, and assume that $c < a$).
>
> a. Model this as a normal-form game.
>
> b. What is the Nash (Cournot) equilibrium of the game in which firms choose their quantities simultaneously?
>
> c. What happens to the equilibrium price as n approaches infinity? Is this familiar?

(a) What is asked. Write the n-firm Cournot game in normal form, find its Nash equilibrium, and find the limit of the price as $n \to \infty$.

(b) Method. Normal-form games with continuous strategies (Tadelis §3.1, pp. 46–50; Lecture 1, slide 13) and Nash equilibrium as the crossing of best responses (Tadelis §5.2.3, pp. 87–88; Lecture 2, slides 20–27, with the n-firm case on slides 26–27).

(c) Solution. Book typo. The statement prints "$Q = q_i + \cdots + q_n$". Aggregate production is the sum over all firms, $Q = q_1 + \cdots + q_n$; the subscript i is a typo for 1, and the solution uses $Q = \sum_{j=1}^{n} q_j$.

Notation. $N = \{1, \ldots, n\}$; $q_i \geq 0$ firm i's output; $Q = \sum_j q_j$; $Q_{-i} = Q - q_i$ the rivals' total; $a > c \geq 0$. The book defines $P(Q) = a - Q$ for $Q \leq a$; for $Q > a$ we take $P = 0$ (Gibbons's convention, p. 15). This only matters off equilibrium: there a firm earns $-c\,q_i \leq 0$.

(a) Normal form. Players N; strategies $S_i = [0, \infty)$; payoffs

$$u_i(q_i, q_{-i}) = \max\{0,\ a - q_i - Q_{-i}\}\, q_i - c\, q_i. \tag{1}$$

(b) Best response. Fix Q_{-i}. Raising q_i beyond $a - Q_{-i}$ only adds cost, so consider $q_i \leq a - Q_{-i}$, where $u_i = (a - c - Q_{-i} - q_i)\, q_i$ is strictly concave ($\partial^2 u_i / \partial q_i^2 = -2$). The first-order condition is

$$a - c - Q_{-i} - 2q_i = 0, \tag{2}$$

and with the constraint $q_i \geq 0$,

$$BR_i(Q_{-i}) = \max\left\{0,\ \frac{a - c - Q_{-i}}{2}\right\}. \tag{3}$$

Every firm produces in equilibrium. Suppose some firm has $q_i^* = 0$. By (3) that requires $Q_{-i}^* \geq a - c$, so some rival j produces $q_j^* > 0$ and, by (2) for j, $q_j^* = a - c - Q^*$ (because

$Q^*_{-j} + 2q^*_j = Q^* + q^*_j$). But $Q^* \geq Q^*_{-i} \geq a - c$ makes $q^*_j \leq 0$, a contradiction. So every firm is interior and (2) holds for all i.

Solving. Rewrite (2) with $Q_{-i} = Q - q_i$:

$$a - c - (Q - q_i) - 2q_i = 0 \implies q_i = a - c - Q. \tag{4}$$

The right side of (4) is the same for every firm, so all firms produce the same amount, $q_i = q$. Summing (4) over the n firms:

$$Q = n(a - c - Q) \implies (n+1)Q = n(a-c) \implies Q^* = \frac{n(a-c)}{n+1}. \tag{5}$$

Substituting (5) into (4):

$$q^*_i = a - c - \frac{n(a-c)}{n+1} = \frac{a-c}{n+1}, \qquad i = 1, \ldots, n. \tag{6}$$

Since $Q^* < a - c < a$, the price is on the linear part of demand:

$$P^* = a - Q^* = a - \frac{n(a-c)}{n+1} = \frac{a+nc}{n+1}, \tag{7}$$

and each firm earns

$$u^*_i = (P^* - c)\, q^*_i = \Big(\frac{a+nc}{n+1} - c\Big)\frac{a-c}{n+1} = \Big(\frac{a-c}{n+1}\Big)^2.$$

The equilibrium is unique: the argument above showed every NE is interior, and (4)–(6) have one solution. In the slides' notation, $q^m = (a-c)/2$ and (6) reads $q^*_i = 2q^m/(n+1)$; $n = 1$ gives the monopoly output q^m and $n = 2$ gives $\frac{2}{3}q^m$ (slide 24).

(c) Limit. From (7),

$$P^* = \frac{a+nc}{n+1} = c + \frac{a-c}{n+1} \xrightarrow[n\to\infty]{} c,$$

and $Q^* \to a - c = q^{cp}$, the competitive output, with each firm's output and profit going to zero. This is the familiar perfectly competitive outcome: price equals marginal cost.

(d) Interpretation. The markup $P^* - c = (a-c)/(n+1)$ falls as $1/(n+1)$. Each firm's output lowers the price for everyone, but a firm counts only the loss on its own units; with more firms its own share is smaller, so it expands more aggressively and the outcome approaches competition (slide 27). Contrast with Bertrand (Lecture 2, slides 29–40), where two firms already price at cost.

(e) Verification. `check_A2.py` solves the n first-order conditions with sympy for $n = 1, \ldots, 6$ and checks (5)–(7) and the profit formula, checks that $P^* - c = (a-c)/(n+1)$ and that the limit is c, and confirms numerically ($a = 100$, $c = 10$, $n = 3$) that no unilateral deviation on a fine grid beats $q^*_i = 22.5$, including the kinked part of demand. No answer in the book.

(f) Common mistakes.

- Treating Q_{-i} as $(n-1)q$ *before* differentiating. Symmetry may be imposed only after the first-order condition is written for a general profile; otherwise firm i seems to move its rivals' output too.
- Assuming the equilibrium is symmetric without proof. The step "$q_i = a - c - Q$ for every i" shows that every equilibrium is symmetric, and the corner argument rules out firms with $q_i = 0$.

- Reading (c) as "each firm's output goes to zero, so the price goes to a". Total output rises to $a - c$, and the price falls to c.

Answer. (a) $N = \{1, \ldots, n\}$, $S_i = [0, \infty)$, $u_i = (a - \sum_j q_j)\, q_i - c\, q_i$ (with price 0 if $\sum_j q_j > a$); the printed "$Q = q_i + \cdots + q_n$" should be $q_1 + \cdots + q_n$. (b) Unique NE $q_i^* = \frac{a-c}{n+1}$, $Q^* = \frac{n(a-c)}{n+1}$, $P^* = \frac{a+nc}{n+1}$, profit $\left(\frac{a-c}{n+1}\right)^2$. (c) $P^* \to c$: the perfectly competitive price (price = marginal cost).

Item 13: Gibbons Problem 1.5

Exam · difficulty 3/5

Statement (verbatim from the list, item 13)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §1.2 Problems, Problem 1.5, p. 49 [PDF p. 57]. Answer in the source: none (Gibbons prints no answers). Reproduced verbatim.

> 1.5. Consider the following two finite versions of the Cournot duopoly model. First, suppose each firm must choose either half the monopoly quantity, $q_m/2 = (a-c)/4$, or the Cournot equilibrium quantity, $q_c = (a-c)/3$. No other quantities are feasible. Show that this two-action game is equivalent to the Prisoners' Dilemma: each firm has a strictly dominated strategy, and both are worse off in equilibrium than they would be if they cooperated. Second, suppose each firm can choose either $q_m/2$, or q_c, or a third quantity, q'. Find a value for q' such that the game is equivalent to the Cournot model in Section 1.2.A, in the sense that (q_c, q_c) is a unique Nash equilibrium and both firms are worse off in equilibrium than they could be if they cooperated, but neither firm has a strictly dominated strategy.

Referred to in the statement. The Cournot model of Section 1.2.A (Gibbons pp. 15–16 [PDF p. 23–24]) (reproduced from the book):

> Let q_1 and q_2 denote the quantities (of a homogeneous product) produced by firms 1 and 2, respectively. Let $P(Q) = a - Q$ be the market-clearing price when the aggregate quantity on the market is $Q = q_1 + q_2$. (More precisely, $P(Q) = a - Q$ for $Q < a$, and $P(Q) = 0$ for $Q \geq a$.) Assume that the total cost to firm i of producing quantity q_i is $C_i(q_i) = cq_i$. That is, there are no fixed costs and the marginal cost is constant at c, where we assume $c < a$. Following Cournot, suppose that the firms choose their quantities simultaneously. [...]
>
> We assume that the firm's payoff is simply its profit. Thus, the payoff $u_i(s_i, s_j)$ in a general two-player game in normal form can be written here as
>
> $$\pi_i(q_i, q_j) = q_i[P(q_i + q_j) - c] = q_i[a - (q_i + q_j) - c].$$

(a) What is asked. Two finite versions of the Cournot duopoly with $P(Q) = a - Q$ and marginal cost c. (i) With only $q_m/2 = (a-c)/4$ and $q_c = (a-c)/3$ available, show the game is a Prisoners' Dilemma. (ii) With a third quantity q' available, find q' such that (q_c, q_c) is the unique equilibrium and both firms are worse off than in cooperation, but no strategy is strictly dominated.

(b) Method. Normal-form games built from the Cournot payoff (Lecture 2, slides 20–25), strict dominance (Lecture 1) and the cartel as a prisoner's dilemma (Lecture 2, slide 25). Measure quantities in units of $a - c$, so $q_m/2 = \frac{1}{4}$, $q_c = \frac{1}{3}$ and the profit of a firm producing q against a rival producing y is

$$f(q, y) = q\,(1 - q - y) \cdot (a - c)^2,$$

since $\pi = (a - q - y - c)\,q$.

(c) Solution. (i) The 2×2 game. Payoffs in units of $(a-c)^2$: $f(\frac{1}{4}, \frac{1}{4}) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$; $f(\frac{1}{3}, \frac{1}{3}) = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$; $f(\frac{1}{4}, \frac{1}{3}) = \frac{1}{4} \cdot \frac{5}{12} = \frac{5}{48}$; $f(\frac{1}{3}, \frac{1}{4}) = \frac{1}{3} \cdot \frac{5}{12} = \frac{5}{36}$.

$1 \setminus 2$	$q_m/2$	q_c
$q_m/2$	$\frac{1}{8}, \frac{1}{8}$	$\frac{5}{48}, \underline{\frac{5}{36}}$
q_c	$\underline{\frac{5}{36}}, \frac{5}{48}$	$\underline{\frac{1}{9}}, \underline{\frac{1}{9}}$

Against $q_m/2$ the quantity q_c pays $\frac{5}{36} > \frac{1}{8}$ ($0.139 > 0.125$); against q_c it pays $\frac{1}{9} > \frac{5}{48}$ ($0.111 > 0.104$). So q_c strictly dominates $q_m/2$ for each firm, (q_c, q_c) is the unique equilibrium (a dominant-strategy equilibrium), and both earn $\frac{1}{9} < \frac{1}{8}$, what they would earn at $(q_m/2, q_m/2)$. Strictly dominated strategy, worse-off equilibrium: a Prisoners' Dilemma.

(ii) Adding q'. The payoff depends on the rival only through its quantity y, so compare strategies as functions of y:

$$f(q_c, y) - f(q_m/2, y) = \frac{5 - 12y}{144}, \qquad f(q_c, y) - f(q', y) = \left(q'^2 - q' + \tfrac{2}{9}\right) + y\left(q' - \tfrac{1}{3}\right).$$

The first is positive iff $y < \frac{5}{12}$. The second, for $q' > \frac{1}{3}$, increases in y, and at $y = \frac{1}{4}$ equals $(q' - \frac{1}{3})(q' - \frac{5}{12})$.

- If $q' < \frac{5}{12}$, every rival quantity available ($\frac{1}{4}$, $\frac{1}{3}$, q') is below $\frac{5}{12}$, so q_c strictly beats $q_m/2$ against all of them: $q_m/2$ is strictly dominated. Excluded.
- If $q' > \frac{5}{12}$, then $(q' - \frac{1}{3})(q' - \frac{5}{12}) > 0$ and every rival quantity is at least $\frac{1}{4}$, so q_c strictly beats q' against all of them: q' is strictly dominated. Excluded.
- So the only candidate is $q' = \frac{5}{12}(a - c)$, where both differences are ≥ 0 with equality at exactly one rival quantity each ($y = \frac{5}{12}$ in the first, $y = \frac{1}{4}$ in the second).

With $q' = \frac{5}{12}$ the payoff table, in units of $(a - c)^2/144$ (row firm's payoff; the game is symmetric), is

	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{5}{12}$
$\frac{1}{4}$	18	15	12
$\frac{1}{3}$	20	16	12
$\frac{5}{12}$	20	15	10

q_c is at least as good as each of the others against every column, but ties with $q_m/2$ against q' ($12 = 12$) and with q' against $q_m/2$ ($20 = 20$), so *no strategy is strictly dominated* (neither by a pure nor by a mixed strategy). The profile (q_c, q_c) is an equilibrium paying 16 each, which is less than the 18 each of the cooperative profile $(q_m/2, q_m/2)$: both are worse off than under cooperation.

A caveat on uniqueness. Because of the two ties just noted, $(q_m/2, q')$ and $(q', q_m/2)$ are also Nash equilibria, in the weak sense: at $(q_m/2, q')$ firm 1 earns 12 and is indifferent between $q_m/2$ and q_c, firm 2 earns 20 and is indifferent between q' and q_c. So the three-quantity game has three equilibria, not one. No value of q' gives all of the properties together: the two arguments above show that any $q' \neq \frac{5}{12}$ makes some strategy strictly dominated, and $q' = \frac{5}{12}$ creates the ties. The value $q' = 5(a - c)/12$ is the one that matches the exercise as far as it can be matched.

Answer. (i) q_c strictly dominates $q_m/2$ for both firms, (q_c, q_c) is the unique equilibrium with profits $(a-c)^2/9$ each, below the $(a-c)^2/8$ each earned at $(q_m/2, q_m/2)$: a Prisoners' Dilemma. (ii) $q' = 5(a - c)/12$ is the only value for which no strategy is strictly dominated; (q_c, q_c) is then an equilibrium that is worse for both than cooperating, but it is not the *only* one (two weak equilibria appear through ties).

(d) Interpretation. The point of the exercise is that the prisoner's-dilemma logic of the cartel (Lecture 2, slide 25) does not need a strictly dominated strategy: the Cournot model of Section 1.2.A has a continuum of quantities, no strategy is dominated, and still the equilibrium is Pareto-dominated. The finite version with $q' = 5(a - c)/12$ keeps that feature, but only at the knife's edge: the third quantity must be neither too small (it would make $q_m/2$ dominated) nor too big (it would be dominated itself).

(e) Verification. `check_L2.py` uses exact fractions: the two payoff tables, $f(q_c, y) - f(q_m/2, y) = 0$ iff $y = \frac{5}{12}$, the factorisation $(q' - \frac{1}{3})(q' - \frac{5}{12})$ at $y = \frac{1}{4}$, and, for every q' on a grid of $1/240$ other than $\frac{1}{4}, \frac{1}{3}, \frac{5}{12}$, that q_c pure-strictly dominates q' (if $q' > \frac{5}{12}$) or $q_m/2$ (if $q' < \frac{5}{12}$). At $q' = \frac{5}{12}$ it checks the table above, that nothing is strictly dominated (linear programming over mixtures) and that the pure equilibria are exactly (q_c, q_c), $(q_m/2, q')$ and $(q', q_m/2)$. Gibbons gives no answer.

(f) Common mistakes.

- Computing profits with the market price $a - q_i - q_j$ but forgetting to subtract c (the factor is $a - c - q_i - q_j$).
- Choosing q' at random and checking only that (q_c, q_c) is an equilibrium: the requirement is also that nothing is strictly dominated.
- Reading "worse off than if they cooperated" as comparing with the monopoly profit: the comparison is with $(q_m/2, q_m/2)$, each earning $(a - c)^2/8$.

Item 14: Gibbons Problem 1.6

Exam · difficulty 3/5

Statement (verbatim from the list, item 14)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §1.2 Problems, Problem 1.6, p. 49 [PDF p. 57]. Answer in the source: none (Gibbons prints no answers). Reproduced verbatim.

> 1.6. Consider the Cournot duopoly model where inverse demand is $P(Q) = a - Q$ but firms have asymmetric marginal costs: c_1 for firm 1 and c_2 for firm 2. What is the Nash equilibrium if $0 < c_i < a/2$ for each firm? What if $c_1 < c_2 < a$ but $2c_2 > a + c_1$?

(a) What is asked. Find the Cournot equilibrium with marginal costs c_1, c_2 when both costs are below $a/2$, and when $2c_2 > a + c_1$.

(b) Method. Cournot equilibrium as the crossing of best responses (Gibbons §1.2.A, pp. 14–21; Tadelis §5.2.3, eq. (5.2), p. 87; Lecture 2, slides 20–24), now with a corner: a best response can be $q_i = 0$.

(c) Solution. Notation. $q_i \geq 0$; $P(Q) = a - Q$ for $Q < a$ and 0 for $Q \geq a$ (the referenced model, Gibbons p. 15); firm i's cost $c_i q_i$, $0 \leq c_i < a$; $j \neq i$.

Best responses. For $q_i < a - q_j$ the payoff is

$$u_i(q_i, q_j) = (a - q_i - q_j - c_i)\, q_i, \tag{1}$$

strictly concave in q_i; for $q_i \geq a - q_j$ the price is zero and $u_i = -c_i q_i \leq 0$. The first-order condition of (1) is $a - c_i - q_j - 2q_i = 0$, so

$$BR_i(q_j) = \max\left\{0,\ \frac{a - c_i - q_j}{2}\right\}. \tag{2}$$

The corner $BR_i = 0$ occurs exactly when $q_j \geq a - c_i$: the rival alone already pushes the price down to c_i or below.

Interior candidate. If both are interior, (2) gives $2q_1 = a - c_1 - q_2$ and $2q_2 = a - c_2 - q_1$. Substituting $q_2 = (a - c_2 - q_1)/2$ into the first:

$$2q_1 = a - c_1 - \left(\frac{a - c_2 - q_1}{2}\right) \implies 4q_1 = 2a - 2c_1 - a + c_2 + q_1 \implies q_1^* = \frac{a - 2c_1 + c_2}{3}, \tag{3}$$

and symmetrically

$$q_2^* = \frac{a - 2c_2 + c_1}{3}. \tag{4}$$

This is a NE exactly when both (3) and (4) are positive.

Case 1: $0 < c_i < a/2$ for $i = 1, 2$. Then $a - 2c_i > 0$, so $a - 2c_i + c_j > c_j > 0$: (3) and (4) are positive and form a NE. No other NE exists. A corner NE with $q_2 = 0$ would need $q_1 = BR_1(0) = (a - c_1)/2$ and, by (2), $q_1 \geq a - c_2$, that is, $(a - c_1)/2 \geq a - c_2 \iff 2c_2 \geq a + c_1$, impossible because $2c_2 < a < a + c_1$. The same holds for $q_1 = 0$, and $(0, 0)$ fails because $BR_i(0) = (a - c_i)/2 > 0$. The unique NE is

$$q_1^* = \frac{a - 2c_1 + c_2}{3}, \quad q_2^* = \frac{a - 2c_2 + c_1}{3}, \quad Q^* = \frac{2a - c_1 - c_2}{3}, \quad P^* = \frac{a + c_1 + c_2}{3},$$

with profits $u_i^* = (P^* - c_i)\, q_i^* = (q_i^*)^2$, because $P^* - c_i = (a + c_1 + c_2 - 3c_i)/3 = (a - 2c_i + c_j)/3 = q_i^*$.

Case 2: $c_1 < c_2 < a$ and $2c_2 > a + c_1$. Now (4) is negative, $a - 2c_2 + c_1 < 0$, so there is no interior NE. Try firm 1 as a monopolist:

$$q_1 = BR_1(0) = \frac{a - c_1}{2}, \qquad q_2 = 0.$$

Firm 2's best response to it, by (2):

$$BR_2\Big(\frac{a - c_1}{2}\Big) = \max\Big\{0,\ \frac{a - c_2 - \frac{a - c_1}{2}}{2}\Big\} = \max\Big\{0,\ \frac{a + c_1 - 2c_2}{4}\Big\} = 0,$$

since $a + c_1 - 2c_2 < 0$. So $(\frac{a - c_1}{2}, 0)$ is a NE. It is the only one: the other corner, $q_1 = 0$ and $q_2 = (a - c_2)/2$, would need $(a - c_2)/2 \geq a - c_1$, that is, $2c_1 \geq a + c_2$, but $2c_1 < c_1 + c_2 < a + c_2$; and $(0, 0)$ fails as before. At the NE the price is the monopoly price of firm 1,

$$P^* = a - \frac{a - c_1}{2} = \frac{a + c_1}{2} < c_2,$$

so firm 2 cannot sell any unit profitably; firm 1 earns $\big(\frac{a - c_1}{2}\big)^2$.

(d) Interpretation. With similar costs both firms produce, and the cheaper one produces more: $q_1^* - q_2^* = c_2 - c_1$. When firm 2's cost exceeds firm 1's monopoly price, $c_2 > (a + c_1)/2$, firm 2 is priced out even though it is not shut out by any rule: firm 1 simply behaves as a monopolist and firm 2's best reply is to stay out. The condition $2c_2 > a + c_1$ is exactly "c_2 above firm 1's monopoly price". (Case 1 is sufficient, not necessary, for the interior NE: (3)–(4) are positive whenever $2c_i < a + c_j$ for both firms, even if some $c_i \geq a/2$.)

(e) Verification. `check_A2.py` derives (2)–(4) with sympy, checks $u_i^* = (q_i^*)^2$, and for random cost pairs in each case finds the NE by iterating the exact best responses (2) and confirms it equals the formulas above; it also checks on a fine grid that no unilateral deviation pays, including outputs that push Q above a. No answer in the book. (Kumar's unofficial Gibbons manual, not in `Livros/`, was not consulted.)

(f) Common mistakes.

- Reporting $q_2^* = (a - 2c_2 + c_1)/3$ in the second case. It is negative there; the formula is a Nash equilibrium only when it is non-negative, and the corner of the best response takes over.
- Dropping the $\max\{0, \cdot\}$ from the best response. Without it the second case has no solution at all.
- Checking only that firm 2 does not want to produce. Uniqueness also needs the other corner and $(0, 0)$ to be ruled out, as done above.

Answer. If $0 < c_i < a/2$: unique NE $q_1^* = \frac{a - 2c_1 + c_2}{3}$, $q_2^* = \frac{a - 2c_2 + c_1}{3}$, price $\frac{a + c_1 + c_2}{3}$, profits $(q_i^*)^2$. If $c_1 < c_2 < a$ and $2c_2 > a + c_1$: unique NE $q_1^* = \frac{a - c_1}{2}$, $q_2^* = 0$; firm 1 is a monopolist at price $\frac{a + c_1}{2} < c_2$.

Item 15: Gibbons Problem 1.7

Exam · difficulty 3/5

Statement (verbatim from the list, item 15)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §1.2 Problems, Problem 1.7, pp. 49–50 [PDF p. 57–58]. Answer in the source: none (Gibbons prints no answers). Reproduced verbatim.

> 1.7. In Section 1.2.B, we analyzed the Bertrand duopoly model with differentiated products. The case of homogeneous products yields a stark conclusion. Suppose that the quantity that consumers demand from firm i is $a - p_i$ when $p_i < p_j$, 0 when $p_i > p_j$, and $(a - p_i)/2$ when $p_i = p_j$. Suppose also that there are no fixed costs and that marginal costs are constant at c, where $c < a$. Show that if the firms choose prices simultaneously, then the unique Nash equilibrium is that both firms charge the price c.

(a) What is asked. Show that in homogeneous-good Bertrand duopoly with unit cost $c < a$ the unique Nash equilibrium is $p_1 = p_2 = c$.

(b) Method. Nash equilibrium (Gibbons §1.1.C, pp. 7–12; Lecture 2, slide 14) in a game with a discontinuous payoff, so there is no first-order condition: existence by checking every deviation, uniqueness by showing that every other profile has a profitable deviation. Gibbons §1.2.B (pp. 21–22) treats the differentiated case; the slides prove this exact result (Lecture 2, slides 29–39, "(c, c) is the unique equilibrium"), and the case split below follows them.

(c) Solution. **The game.** Players $1, 2$; strategies $p_i \in [0, \infty)$. Demand for firm i is

$$D_i(p_i, p_j) = \begin{cases} a - p_i & p_i < p_j, \\ \frac{1}{2}(a - p_i) & p_i = p_j, \\ 0 & p_i > p_j, \end{cases}$$

read as $\max\{0, \cdot\}$ when $p_i > a$ (no negative sales), and profit is $\pi_i = (p_i - c)\, D_i(p_i, p_j)$, with $0 \le c < a$.

Step 1: (c, c) is a Nash equilibrium. At (c, c) each firm sells $(a - c)/2$ at margin 0 and earns 0. Fix $p_j = c$ and let firm i deviate.

- $p_i > c$: $D_i = 0$, so $\pi_i = 0$. Not better.
- $p_i < c$: firm i takes the whole market, $D_i = a - p_i > 0$ (because $p_i < c < a$), at a negative margin: $\pi_i = (p_i - c)(a - p_i) < 0$. Worse.

No deviation is profitable, so (c, c) is a Nash equilibrium.

Step 2: no other profile is a Nash equilibrium. Take $(p_1, p_2) \neq (c, c)$ and label the firms so that $p_1 \le p_2$ (the game is symmetric). Every such profile falls in exactly one case.

(i) $p_1 < c$. Firm 1 sells a positive amount ($a - p_1$ or $(a - p_1)/2$, both > 0 because $p_1 < c < a$) at a negative margin, so $\pi_1 < 0$. Deviating to any $p_1' > p_2$ gives $D_1 = 0$ and $\pi_1 = 0 > \pi_1$. Not NE.

(ii) $p_1 = c < p_2$. Firm 1 earns 0. Let $\bar{p} = \min\{p_2, a\} > c$ and deviate to $p_1' = (c + \bar{p})/2 \in (c, \bar{p})$: still below p_2, so firm 1 keeps the whole market, $D_1 = a - p_1' > 0$, and $\pi_1 = (p_1' - c)(a - p_1') > 0$. Not NE.

(iii) $c < p_1 < p_2$. Firm 2 sells nothing and earns 0. Let $\bar{p} = \min\{p_1, a\} > c$ and let firm 2 deviate to $p_2' = (c + \bar{p})/2 \in (c, \bar{p})$: it undercuts firm 1, sells $a - p_2' > 0$ at a positive margin, $\pi_2 > 0$. Not NE.

(iv) $c < p_1 = p_2 = p$. If $p \geq a$, demand is 0 and both earn 0; firm 1 deviating to $(c + a)/2$ sells $(a-c)/2 > 0$ at margin $(a-c)/2 > 0$. Not NE. If $c < p < a$, each earns $\pi_1 = \frac{1}{2}(p-c)(a-p) > 0$. Firm 1 undercuts to the midpoint $p' = (p + c)/2$, which is in (c, p), takes the whole market and earns

$$(p' - c)(a - p') = \frac{p - c}{2}\Big(a - p + \frac{p - c}{2}\Big) = \frac{1}{2}(p - c)(a - p) + \frac{(p - c)^2}{4} > \frac{1}{2}(p - c)(a - p).$$

Not NE.

The cases cover every profile with $p_1 \leq p_2$ other than (c, c): $p_1 < c$ is (i); $p_1 = c$ forces $p_2 > c$, case (ii); $p_1 > c$ is (iii) or (iv). With Step 1, (c, c) is the unique Nash equilibrium. ■

(d) Interpretation. With identical products and constant marginal cost, whoever is cheaper takes the whole market, so any price above cost invites a slight undercut that roughly doubles the undercutter's sales. Competition drives price to marginal cost and profits to zero with only two firms: the *Bertrand paradox* (Lecture 2, slide 39), in sharp contrast with Cournot, where two firms keep a markup of $(a - c)/3$ (the n-firm exercise, Tadelis 5.8). The result rests on the continuous price grid, unlimited capacity and homogeneous goods; relaxing any of them (a smallest currency unit, capacity constraints, differentiation as in Gibbons §1.2.B) restores positive margins.

(e) Verification. `check_A2.py` works in exact rational arithmetic with $a = 10$, $c = 2$: it checks that no deviation on a fine price grid (including prices above a) improves on (c, c), and, for 3,000 random profiles $(p_1, p_2) \neq (c, c)$, that the deviation named in the case that applies is strictly profitable. Gibbons gives no answer.

(f) Common mistakes.

- Using a first-order condition. Profit jumps at $p_i = p_j$, so the best response does not exist when $p_j > c$ (one always wants to be slightly below p_j); the argument must be by deviations.
- Forgetting case (ii): $p_1 = c < p_2$ looks "competitive", but firm 1 can raise its price toward p_2 and earn a positive profit.
- Undercutting by "ε" without checking the gain. The midpoint deviation in (iv) shows explicitly that the undercutter earns more than its half of the market.

Answer. At (c, c) both earn 0; raising the price loses all sales and cutting it sells at a loss, so no deviation pays. Any other profile has a profitable deviation: a firm below cost raises its price above the rival's; a firm at cost facing a higher rival price raises its price toward the rival's; a firm priced out undercuts; and at a common price above cost a firm undercuts slightly and nearly doubles its profit. Hence $p_1 = p_2 = c$ is the unique Nash equilibrium.

Item 16: Tadelis Exercise 6.8

Exam · difficulty 4/5

Statement (verbatim from the list, item 16)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 6.8 "Market Entry", p. 125 [PDF p. 144]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim.

> **Market Entry:** Three firms are considering entering a new market. The payoff for each firm that enters is $\frac{150}{n}$, where n is the number of firms that enter. The cost of entering is 62.
>
> a. Find all the pure-strategy Nash equilibria.
>
> b. Find the symmetric mixed-strategy equilibrium in which all three players enter with the same probability.

(a) What is asked. Find all pure Nash equilibria of the three-firm entry game, and the symmetric mixed equilibrium.

(b) Method. Pure NE by checking unilateral deviations (Tadelis Def. 5.1, p. 80); mixed NE by the principle of indifference (Tadelis Prop. 6.1, p. 108; Lecture 2, slides 44–46), here with three players.

(c) Solution. Notation. $I = \{1, 2, 3\}$, $S_i = \{E, O\}$ (enter, stay out). With n entrants an entrant's payoff is $\pi(n) = 150/n - 62$; staying out pays 0:

$$\pi(1) = 150 - 62 = 88, \qquad \pi(2) = 75 - 62 = 13, \qquad \pi(3) = 50 - 62 = -12.$$

(a) Pure equilibria. Classify profiles by the number n of entrants.

- $n = 0$: any firm that enters gets $\pi(1) = 88 > 0$. Not NE.
- $n = 1$: an outsider that enters gets $\pi(2) = 13 > 0$. Not NE.
- $n = 2$: each entrant gets $13 > 0$, so it does not want to leave; the outsider would get $\pi(3) = -12 < 0$ by entering, so it stays out. NE.
- $n = 3$: each gets $-12 < 0$ and would rather stay out. Not NE.

The pure NE are the three profiles in which exactly two firms enter: (E, E, O), (E, O, E), (O, E, E), each with payoffs 13 to the entrants and 0 to the outsider.

(b) Symmetric mixed equilibrium. Let each firm enter with probability p. The pure symmetric profiles are not NE (by (a), $p = 0$ and $p = 1$ fail), so $p \in (0, 1)$ and every firm must be indifferent between E and O. Fix firm i. The number of other entrants is 0, 1 or 2 with probabilities $(1-p)^2$, $2p(1-p)$, p^2, so

$$u_i(E, p) = (1-p)^2 \cdot 88 + 2p(1-p) \cdot 13 + p^2 \cdot (-12), \qquad u_i(O, p) = 0. \tag{1}$$

Expanding the first expression in (1):

$$88(1 - 2p + p^2) + 26(p - p^2) - 12p^2 = 88 - 176p + 88p^2 + 26p - 26p^2 - 12p^2 = 88 - 150p + 50p^2.$$

Indifference, $u_i(E,p)=0$, gives

$$50p^2-150p+88=0 \iff 25p^2-75p+44=0,$$

$$p=\frac{75\pm\sqrt{75^2-4\cdot 25\cdot 44}}{2\cdot 25}=\frac{75\pm\sqrt{5625-4400}}{50}=\frac{75\pm 35}{50}\in\left\{\frac{4}{5},\ \frac{11}{5}\right\}.$$

Only $p=4/5$ is a probability. The symmetric mixed NE has each firm entering with probability $p^*=\frac{4}{5}$, and each firm's expected payoff is 0 (the payoff of staying out).

Check: at $p=0.8$, $(0.2)^2\cdot 88+2(0.8)(0.2)\cdot 13+(0.8)^2\cdot(-12)=3.52+4.16-7.68=0$. Why only interior p can work: $u_i(E,p)=88-150p+50p^2$ is positive for $p<\frac{4}{5}$ (every firm would enter for sure) and negative for $p\in(\frac{4}{5},1]$ (every firm would stay out), so $p^*=\frac{4}{5}$ is the only symmetric equilibrium.

The number of entrants at $p^*=\frac{4}{5}$. It is binomial with $n=3$: $\Pr(0)=0.2^3=0.008$, $\Pr(1)=3(0.8)(0.2)^2=0.096$, $\Pr(2)=3(0.8)^2(0.2)=0.384$, $\Pr(3)=0.8^3=0.512$; the expected number of entrants is $3p^*=2.4$.

Beyond what is asked: an asymmetric mixed equilibrium. Let firm 1 enter for sure and firms 2 and 3 enter with probability r each. Firm 2 faces one sure entrant and firm 3 mixing, so it is indifferent when $(1-r)\cdot 13+r\cdot(-12)=0$, i.e. $r=\frac{13}{25}$; the same holds for firm 3. Firm 1 must prefer entering: $(1-r)^2\cdot 88+2r(1-r)\cdot 13+r^2\cdot(-12)=88-150r+50r^2=88-78+13.52=23.52>0$ at $r=0.52$. So firm 1 entering and firms 2, 3 entering with probability $\frac{13}{25}$ (and the two permutations) are also Nash equilibria.

(d) Interpretation. Entry is a coordination problem over *who* enters: the market supports two firms, not three. The pure equilibria solve it but need the firms to know who stays out. Without such a convention, the symmetric equilibrium has each firm entering with probability 0.8. Then all three enter with probability $0.8^3=0.512$ and lose money, and the expected profit of the industry is zero: all the rents of the two-firm market are dissipated by miscoordination.

(e) Verification. `check_A2.py` brute-forces the 2^3 profiles (pure NE are exactly the three two-entrant profiles), solves the indifference condition with sympy (roots $\frac{4}{5}$ and $\frac{11}{5}$), and checks that at $p=\frac{4}{5}$ entering and staying out both pay 0 while $u_i(E,p)$ is positive for $p<\frac{4}{5}$ and negative for $p\in(\frac{4}{5},1]$. No answer in the book.

(f) Common mistakes.

- Using a binomial over *three* firms in the indifference condition. Firm i's payoff depends on how many of the *other two* enter; the firm's own choice is what is being compared.
- Keeping the root $p=\frac{11}{5}$, or discarding $\frac{4}{5}$ because the expected payoff is zero. A zero expected payoff is exactly what indifference with staying out requires.
- Concluding that the symmetric equilibrium is the only mixed one. The question restricts attention to it; asymmetric mixed equilibria exist (see above).

Answer. (a) The three profiles with exactly two entrants, (E,E,O), (E,O,E), (O,E,E): each entrant earns $75-62=13$, and the outsider would earn $50-62=-12$ by entering. (b) Each firm enters with probability $p^*=\frac{4}{5}$ (from $50p^2-150p+88=0$); expected payoff 0.

Item 17: MWG Exercise 8.D.2

Advanced · difficulty 4/5

Statement (verbatim from the list, item 17)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 Exercises, Exercise 8.D.2ˆB, p. 263 [PDF p. 142]. Answer in the source: none in the book (the MWG solutions manual covers Chapter 8). Reproduced verbatim.

> Show that if there is a unique profile of strategies that survives iterated removal of strictly dominated strategies, this profile is a Nash equilibrium.

(a) What is asked. Prove that if exactly one strategy profile survives iterated removal of strictly dominated strategies, that profile is a Nash equilibrium.

(b) Method. Iterated deletion of strictly dominated strategies (MWG §8.B, pp. 236–241; Lecture 1, slide 29) and Nash equilibrium (MWG Def. 8.D.1, pp. 246–247; Lecture 2, slide 14). The proof is the slides' Proposition 2 (Lecture 2, slide 17): follow a chain of dominating strategies, along which the payoff against the survivor's profile strictly rises, until it reaches the surviving strategy. The game is finite (MWG §8.B). The same chain works whether the dominators are pure strategies (MWG Def. 8.B.2) or mixed (Def. 8.B.4); both versions are proved.

(c) Solution. Setting. $\Gamma = [I, \{S_i\}, \{u_i\}]$, every S_i finite. $S^0 = S$, and S^{k+1} removes from S^k every strategy strictly dominated given S^k (by a pure strategy in S_i^k, or by a mixed strategy $\sigma_i \in \Delta(S_i^k)$). The process stops after finitely many rounds at S^∞. Hypothesis: $S^\infty = \{s^*\}$.

Claim. For every player i and every $s_i \in S_i$, $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$.

Step 1: one link of the chain. Let $s_i \neq s_i^*$. Then $s_i \notin S_i^\infty$, so s_i was deleted in some round, say round $k+1$: there is a dominator, pure $t_i \in S_i^k$ or mixed $\sigma_i \in \Delta(S_i^k)$, with

$$u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i}) \quad \text{for every } s_{-i} \in S_{-i}^k. \tag{1}$$

Since s^* is never deleted, $s_{-i}^* \in S_{-i}^k$, and (1) holds at $s_{-i} = s_{-i}^*$:

$$\sum_{t_i \in S_i^k} \sigma_i(t_i)\, u_i(t_i, s_{-i}^*) = u_i(\sigma_i, s_{-i}^*) > u_i(s_i, s_{-i}^*).$$

A weighted average exceeds $u_i(s_i, s_{-i}^*)$ only if at least one of its terms does, so there is a pure t_i with $\sigma_i(t_i) > 0$ (in the pure case, t_i itself) such that

$$t_i \in S_i^k \quad \text{and} \quad u_i(t_i, s_{-i}^*) > u_i(s_i, s_{-i}^*). \tag{2}$$

Call t_i a *successor* of s_i. Since $t_i \in S_i^k$, it survived at least until the round in which s_i was deleted.

Step 2: the chain. Start at any $s_i^{(0)} = s_i \neq s_i^*$. While the current strategy is not s_i^*, it was deleted at some round, so by Step 1 it has a successor; let $s_i^{(m+1)}$ be a successor of $s_i^{(m)}$. By (2),

$$u_i(s_i^{(0)}, s_{-i}^*) < u_i(s_i^{(1)}, s_{-i}^*) < u_i(s_i^{(2)}, s_{-i}^*) < \cdots$$

The payoffs strictly increase, so no strategy appears twice; S_i is finite, so the chain cannot go on forever. It stops only at a strategy with no successor, that is, at a strategy that is never deleted: s_i^*, the only one in S_i^∞. Hence

$$u_i(s_i, s_{-i}^*) < u_i(s_i^*, s_{-i}^*) \quad \text{for every } s_i \neq s_i^*.$$

Step 3: conclusion. No player gains by a unilateral pure deviation from s^*, so s^* is a Nash equilibrium (in fact a *strict* one). Mixed deviations do not help either, since $u_i(\sigma_i, s^*_{-i})$ is a weighted average of the pure payoffs, each at most $u_i(s^*)$. ■

Remarks.

- *Uniqueness.* Every Nash equilibrium survives IESDS (Lecture 2, Proposition 1, slide 16), so s^* is also the only pure Nash equilibrium; and in any mixed equilibrium every strategy in the support survives (MWG manual's argument below), so s^* is the only Nash equilibrium of any kind.
- *Finiteness* is used in Step 2: in an infinite game the chain of successors need not end at s^*, and a unique survivor need not be an equilibrium. (Outside the course books: Dufwenberg and Stegeman, *Econometrica* 2002, study this.) Infinite games in the course where IESDS has a unique survivor, such as the Cournot duopoly of MWG 8.B.5, satisfy the conclusion by direct computation.

(d) Interpretation. If common knowledge of rationality alone pins down a single profile, that profile is self-enforcing: no player can do better against what the others are predicted to do. Nash equilibrium is weaker than IESDS in what it assumes about beliefs, and this result says the two agree whenever IESDS is decisive, as in the campaigning game (Tadelis 4.7) or the discrete auction (Tadelis 4.3).

(e) Verification. `check_A2.py` generates random finite games (two players, two to four strategies, integer payoffs) until it has 1,000 with a unique IESDS survivor under pure dominance, and checks that each survivor is a strict Nash equilibrium and the only pure one; it also builds the chain of Step 2 explicitly in each game and checks that payoffs rise along it and that it ends at s^*_i. *Comparison with the MWG solutions manual* (p. 8-9): the manual proves the converse inclusion, that every (mixed) Nash equilibrium survives iterated strict dominance, and concludes that the equilibrium must be the unique survivor. That route needs an equilibrium to exist in the first place, which in a finite game is Nash's theorem (MWG Prop. 8.D.2) but is not stated in the manual's answer. The direct chain argument above needs no existence theorem. The conclusions agree.

(f) Common mistakes.

- Proving the converse (every Nash equilibrium survives IESDS) and stopping. That shows the survivor is the only *candidate*; existence of an equilibrium is a separate step.
- Assuming the strategy that dominated s_i survives to the end. It may itself be deleted later; the chain handles this.
- Comparing payoffs against all s_{-i} instead of against s^*_{-i}. Only the inequality at s^*_{-i} is needed, and it holds because s^*_{-i} is never deleted.

Answer. Let s^* be the unique survivor and $s_i \neq s^*_i$. When s_i was deleted, its dominator (or a pure strategy in its support) t_i was still present and, since s^*_{-i} was also present, $u_i(t_i, s^*_{-i}) > u_i(s_i, s^*_{-i})$. Repeating from t_i gives a chain with strictly rising payoffs against s^*_{-i}; in a finite game it ends at a never-deleted strategy, s^*_i. So $u_i(s^*_i, s^*_{-i}) > u_i(s_i, s^*_{-i})$ for all $s_i \neq s^*_i$ and all i: s^* is a (strict) Nash equilibrium.

Item 18: MWG Exercise 8.C.1

Advanced · difficulty 3/5

Statement (verbatim from the list, item 18)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 Exercises, Exercise 8.C.1ˆA, p. 262 [PDF p. 142]. Answer in the source: none in the book (the MWG solutions manual covers Chapter 8). Reproduced verbatim.

> Argue that if elimination of strictly dominated strategies yields a unique prediction in a game, this prediction also results from eliminating strategies that are never a best response.

(a) What is asked. Argue that if iterated elimination of strictly dominated strategies leaves a unique prediction, iterated elimination of strategies that are never a best response leaves the same one.

(b) Method. A two-inclusions argument. A strictly dominated strategy is never a best reply, so never-best-response elimination removes *at least* what IESDS removes, round by round; and elimination of never-best-replies can never remove everything, because a best reply exists in a finite game. Definitions: iterated strict dominance (MWG §8.B) and never-best-response elimination, MWG Definitions 8.C.1–8.C.2 (rationalizable strategies; MWG §8.C, pp. 242–245); the slides use the same idea for rationalizability (Lecture 2, slides 11–13). The game is finite.

(c) Solution. **Notation.** $D_i^0 = R_i^0 = S_i$. In round k,

- D_i^k = strategies in D_i^{k-1} not strictly dominated (by a pure or mixed strategy over D_i^{k-1}) against every $s_{-i} \in D_{-i}^{k-1}$: iterated strict dominance;
- R_i^k = strategies in R_i^{k-1} that are a best reply, among all of S_i, to some belief over R_{-i}^{k-1}: iterated removal of never-best-replies, whose limit R^∞ is the set of rationalizable strategies (MWG Definition 8.C.2).

Let $D^\infty = \bigcap_k D^k$ and $R^\infty = \bigcap_k R^k$; suppose $D^\infty = \{s^*\}$.

Step 1: $R^k \subseteq D^k$ for every k. By induction. True for $k = 0$. Suppose $R^{k-1} \subseteq D^{k-1}$, and take $s_i \in R_i^k$ with $s_i \notin D_i^k$. Then s_i is strictly dominated by some σ_i against every $s_{-i} \in D_{-i}^{k-1}$, hence against every belief supported on D_{-i}^{k-1} (an expectation of strictly positive differences is strictly positive). Beliefs over R_{-i}^{k-1} are supported on $R_{-i}^{k-1} \subseteq D_{-i}^{k-1}$, so against every belief over R_{-i}^{k-1} the strategy s_i is strictly worse than σ_i and hence never a best reply. That contradicts $s_i \in R_i^k$. So $R_i^k \subseteq D_i^k$. Taking the limit, $R^\infty \subseteq D^\infty$.

Step 2: $R^\infty \neq \emptyset$. Each R_i^k is non-empty by induction: given $R_{-i}^{k-1} \neq \emptyset$, take any belief over it; a best reply in S_i exists because S_i is finite, and it lies in R_i^{k-1} (if it had been deleted in an earlier round $m \leq k-1$, it would not be a best reply to any belief over $R_{-i}^{m-1} \supseteq R_{-i}^{k-1}$). So it lies in R_i^k. The sets are finite, nested and non-empty, hence their intersection R^∞ is non-empty.

Step 3: conclusion. $\emptyset \neq R^\infty \subseteq D^\infty = \{s^*\}$ forces $R^\infty = \{s^*\}$: elimination of never-best-replies yields the same unique prediction. ■

Answer. Every strictly dominated strategy is never a best reply, so round by round never-best-reply elimination removes at least what IESDS removes: $R^k \subseteq D^k$. Since a best reply always exists, R^∞ is non-empty. If IESDS leaves only s^*, then $\emptyset \neq R^\infty \subseteq \{s^*\}$, so the rationalizable set is also $\{s^*\}$.

(d) Interpretation. Both procedures encode "rational players who know that the others are rational" (Lecture 2, slide 7 for IESDS, slides 11–13 for rationalizability), and the exercise says that the second is never weaker. For two players the two sets are in fact equal (a strategy is never a best reply iff it is strictly dominated, Pearce's lemma, noted in MWG §8.C), so the content is in games with three or more players with independent beliefs, where a strategy can be never a best reply without being strictly dominated and rationalizability can be strictly sharper. Together with exercise 5.1 this gives the chain

strict dominant-strategy eq. ⇒ unique IESDS survivor ⇒ unique rationalizable profile ⇒ unique Nash equilibrium.

(e) Verification. `check_L2.py` computes both procedures on 500 random finite two-player games (strategy counts 2–4, payoffs in $0, \ldots, 6$): IESDS with mixed dominators by linear programming, and rationalizability by a feasibility LP over mixed beliefs. In every game $R^\infty \subseteq D^\infty$ and R^∞ is non-empty; whenever D^∞ is a single profile (more than 30 games) R^∞ is the same profile. (In two-player games they coincide, as the interpretation says.) MWG's solutions manual covers this exercise; this argument is self-contained.

(f) Common mistakes.

- Claiming the two sets are equal in general: with three or more players and independent beliefs, R^∞ can be strictly smaller than D^∞.
- Forgetting Step 2: "$R^\infty \subseteq \{s^*\}$" does not give equality unless R^∞ is non-empty, and non-emptiness is where finiteness (existence of a best reply) is used.
- Letting dominance be pure-only: with mixed dominators the set D^∞ is the right one; with pure dominators the inclusion $R \subseteq D$ still holds, but D is larger.

Item 19: MWG Exercise 8.D.3

Advanced · difficulty 4/5

Statement (verbatim from the list, item 19)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 Exercises, Exercise 8.D.3ˆB, p. 263 [PDF p. 142]. Answer in the source: none in the book (the MWG solutions manual covers Chapter 8). Reproduced verbatim.

> Consider a first-price sealed-bid auction of an object with two bidders. Each bidder i's valuation of the object is v_i, which is known to both bidders. The auction rules are that each player submits a bid in a sealed envelope. The envelopes are then opened, and the bidder who has submitted the highest bid gets the object and pays the auctioneer the amount of his bid. If the bidders submit the same bid, each gets the object with probability $\frac{1}{2}$. Bids must be in dollar multiples (assume that valuations are also).
>
> **(a)** Are any strategies strictly dominated?
>
> **(b)** Are any strategies weakly dominated?
>
> **(c)** Is there a Nash equilibrium? What is it? Is it unique?

(a) What is asked. In a first-price sealed-bid auction with two bidders with known valuations v_1, v_2 and bids in whole dollars: (a) is any strategy strictly dominated? (b) is any weakly dominated? (c) is there a Nash equilibrium, what is it, and is it unique?

(b) Method. Dominance (Lecture 1) and Nash equilibrium for a game with discrete strategies and discontinuous payoffs (Lecture 2, slide 14). The undercutting logic is the one of the Bertrand price game on a grid (Lecture 2, slides 29–39; exercise 5.12): whoever wins by a small margin has no reason to bid more than one unit above the loser. Assume that v_1, v_2 are positive integers and that bids are the non-negative integers, with no upper limit.

(c) Solution. **Payoffs.** For bidder i with bid b_i against b_j:

$$u_i(b_i, b_j) = \begin{cases} v_i - b_i & b_i > b_j, \\ \frac{1}{2}(v_i - b_i) & b_i = b_j, \\ 0 & b_i < b_j. \end{cases}$$

(a) No strategy is strictly dominated. Take any bid b and any (pure or mixed) strategy σ, and let N exceed b and every bid in the support of σ. Against a rival bid of N both b and every bid of σ lose, so each earns 0: σ does not do *strictly* better than b against N. Hence no bid, not even one above the valuation, is strictly dominated.

(b) Weak dominance.

- *Every bid $b \geq v_i$ is weakly dominated by the bid $v_i - 1$* (a legal bid because $v_i \geq 1$). Compare by the rival's bid y: if $y < v_i - 1$, bidding $v_i - 1$ wins for $1 > 0 \geq v_i - b$; if $y = v_i - 1$ it ties for $\frac{1}{2} > 0 \geq v_i - b$ (the bid b wins at a non-positive surplus); if $y > v_i - 1$ it loses and earns 0, while b earns $v_i - b \leq 0$, $\frac{1}{2}(v_i - b) \leq 0$ or 0. So $v_i - 1$ is never worse and is strictly better when $y \leq v_i - 1$.
- *For $v_i \geq 2$ the bid 0 is weakly dominated by the bid 1:* against $y = 0$, bid 1 gives $v_i - 1 \geq \frac{1}{2}v_i$; against $y = 1$ it gives $\frac{1}{2}(v_i - 1) > 0$; against $y \geq 2$ both give 0.

Bidding one's valuation or more is never sensible: at best it wins for no surplus. Bids in $\{1, \ldots, v_i - 1\}$ are not weakly dominated by any other bid (checked by brute force below).

(c) Nash equilibria. Equilibrium profiles split into ties and wins, and each is characterised by the one-unit deviations.

Ties, $b_1 = b_2 = b$. Bidder i earns $\frac{1}{2}(v_i - b)$. Raising the bid by one wins for $v_i - b - 1$, which is not better iff $v_i - b \leq 2$. Lowering loses and gives 0, which is not better iff $b \leq v_i$. Larger changes are worse. So a tie is an equilibrium iff

$$\max\{v_1, v_2\} - 2 \;\leq\; b \;\leq\; \min\{v_1, v_2\} \quad (b \geq 0).$$

Bidder 1 wins, $b_1 > b_2$. If $b_1 > b_2 + 1$, bidder 1 gains by lowering to $b_2 + 1$ ($v_1 - b_2 - 1 > v_1 - b_1$), so $b_1 = b_2 + 1$. Lowering to a tie gives $\frac{1}{2}(v_1 - b_2) = \frac{1}{2}(v_1 - b_1 + 1)$, not better than $v_1 - b_1$ iff $b_1 \leq v_1 - 1$. The loser earns 0; by tying at b_1 he would earn $\frac{1}{2}(v_2 - b_1)$, not better iff $b_1 \geq v_2$ (bidding higher is worse still). With $b_2 = b_1 - 1 \geq 0$ this gives the equilibria

$$(b_1, b_2) = (k, k-1), \qquad \max\{v_2, 1\} \leq k \leq v_1 - 1.$$

Bidder 2 wins: symmetrically $(b_1, b_2) = (k-1, k)$ with $\max\{v_1, 1\} \leq k \leq v_2 - 1$.

Examples. $v_1 = v_2 = 5$: ties at $b \in \{3, 4, 5\}$; no wins. $v_1 = 7$, $v_2 = 5$: tie at $(5, 5)$ and wins $(5, 4)$, $(6, 5)$. $v_1 = 8$, $v_2 = 5$: wins $(5, 4)$, $(6, 5)$, $(7, 6)$ only.

Existence and uniqueness. An equilibrium always exists (for $v_1 \geq v_2$ either $v_1 - v_2 \leq 2$ and the tie range or a win is non-empty, or $v_1 - v_2 \geq 3$ and the win range $[v_2, v_1 - 1]$ is non-empty), but *it is never unique*: with $d = |v_1 - v_2|$ there are 3 equilibria if $0 \leq d \leq 2$ except when $v_1 = v_2 = 1$ (two: $(0, 0)$ and $(1, 1)$), and d equilibria if $d \geq 3$. In every equilibrium bids are at most $\max\{v_1, v_2\}$.

Answer. (a) No strategy is strictly dominated (against a very high rival bid all bids lose). (b) Every bid $b_i \geq v_i$ is weakly dominated by $v_i - 1$ (and the bid 0 by 1 when $v_i \geq 2$). (c) There are Nash equilibria, but never a unique one: ties $b_1 = b_2 = b$ with $\max\{v_1, v_2\} - 2 \leq b \leq \min\{v_1, v_2\}$, and profiles where one bidder wins by exactly one dollar, $b_{\text{win}} = b_{\text{lose}} + 1$ with $\max\{v_{\text{lose}}, 1\} \leq b_{\text{win}} \leq v_{\text{win}} - 1$. The bidder with the higher valuation wins whenever $v_1 - v_2 \geq 3$.

(d) Interpretation. The auction is a Bertrand game in reverse: bidders "undercut" the object's price upward until the surplus is exhausted. With a one-dollar grid the competition stops at a one-dollar margin; the equilibrium price ends between the loser's value and the winner's value minus one, and ties appear when valuations are close. Bidding above value is weakly dominated, which is the reason the first-price auction is not strategy-proof: the *amount* of shading below v_i is what the equilibrium determines (Lecture 7 will return to this with incomplete information).

(e) Verification. `check_L2.py` computes all Nash equilibria by brute force (exact fractions, bids $0, \ldots, \max\{v_1, v_2\} + 4$) for every pair $1 \leq v_1, v_2 \leq 10$ and compares them with the characterisation above and with the counting rule: they coincide in all 100 cases, and the number of equilibria is at least two in each. It also verifies (a) by linear programming against a rival bid above the grid, and (b) by enumerating weak dominance among pure bids for $v = 1, \ldots, 8$ (undominated: $\{1, \ldots, v - 1\}$, and $\{0\}$ if $v = 1$). The MWG solutions manual treats this exercise; the derivation here is complete without it.

(f) Common mistakes.

- Claiming the answer to (a) is "bids above v_i": against a very high rival bid every bid earns zero, so strict dominance fails; they are only weakly dominated.

- Reporting the single equilibrium $b_1 = b_2 = v_2$ (the continuous-bid intuition): on a dollar grid a tie is an equilibrium only if $|v_1 - v_2| \leq 2$, and winning equilibria exist as well.
- Treating a tie as winning with probability one: a tie pays $\frac{1}{2}(v_i - b)$.

Item 20: Tadelis Exercise 5.12

Advanced · difficulty 4/5

Statement (verbatim from the list, item 20)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 5.12 "Asymmetric Bertrand", p. 98 [PDF p. 117]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim.

> **Asymmetric Bertrand:** Consider the Bertrand game with $c_1(q_1) = q_1$ and $c_2(q_2) = 2q_2$ and demand equal to $p = 100 - q$, in which firms must choose prices in increments of \$0.01. We have seen in Section 5.2.4 that one possible Nash equilibrium is $(p_1^*, p_2^*) = (1.99, 2.00)$.
>
> a. Show that there are other Nash equilibria for this game.
>
> b. How many Nash equilibria does this game have?

Referred to in the statement. The Bertrand game of Tadelis §5.2.4 (pp. 88–91 [PDF p. 107–110]) (reproduced from the book):

> Demand is given by $p = 100 - q$; each firm posts a price p_i for otherwise identical goods, all buyers buy from the firm whose price is lowest, and if there is a tie the market splits equally between the two firms. In the asymmetric example $c_1 = 1$ and $c_2 = 2$ and prices are limited to increments of $\epsilon = 0.01$; the book then says that "one possible Nash equilibrium (you are asked to find more in exercise 5.12 at the end of this chapter) is $(p_1, p_2) = (1.99, 2.00)$."

(a) What is asked. In the Bertrand game with $c_1(q_1) = q_1$, $c_2(q_2) = 2q_2$, demand $p = 100 - q$ and prices in increments of $\varepsilon = \$0.01$: (a) show there are Nash equilibria other than $(1.99, 2.00)$; (b) how many Nash equilibria are there?

(b) Method. Bertrand competition with differentiated costs (Lecture 2, slides 29–39; Tadelis §5.2.4, pp. 88–91), where the proof of uniqueness of $p_1 = p_2 = c$ used undercutting; here the undercut is by one grid step. Work case by case on the order of the two prices, giving for each profile either a profitable deviation or a proof that none exists. Prices are multiples of ε in $[0, \infty)$; a price of 100 or more sells nothing.

(c) Solution. Payoffs. Let $D_i = 100 - p_i$ if $p_i < p_j$, $\frac{1}{2}(100 - p_i)$ if $p_i = p_j$ and 0 if $p_i > p_j$ (zero too if $p_i \geq 100$). Then $\pi_1 = (p_1 - 1)D_1$ and $\pi_2 = (p_2 - 2)D_2$.

Lemma (undercutting beats a tie). For $1.02 \leq p < 100$, a firm with marginal cost 1 tied at p earns strictly more by cutting to $p - \varepsilon$. Write $m = p - 1$ and $d = 100 - p > 0$. The tie pays $\frac{1}{2}md$; the undercut pays $(m - \varepsilon)(d + \varepsilon)$; the difference is

$$(m - \varepsilon)(d + \varepsilon) - \tfrac{1}{2}md = \Big(\tfrac{m}{2} - \varepsilon\Big)d + \varepsilon(m - \varepsilon) > 0,$$

since $m \geq 2\varepsilon$ makes the first term non-negative and $m > \varepsilon$ makes the second positive.

Profitable deviations, by case.

- $p_1 = p_2 = p$. If $p < 2$, firm 2 sells $\frac{1}{2}(100 - p) > 0$ at a negative margin; raising its price above p gives 0, better. If $2 \leq p < 100$, firm 1 undercuts, strictly better by the Lemma. If $p \geq 100$ firm 1 earns 0 and gains by charging 50.50. Never an equilibrium.

- $p_1 > p_2$. Firm 1 sells nothing. If $p_2 < 2$, firm 2 sells $100 - p_2 > 0$ at margin $p_2 - 2 < 0$; raising its price above p_1 earns 0, better. If $p_2 \geq 2$, firm 1 charges $\min\{p_2 - \varepsilon, 50.50\}$, which wins the market at a margin of at least 0.99, a strictly positive profit. Never an equilibrium.

$p_1 < p_2$**: the equilibria.** Firm 2 sells nothing and earns 0.

- *Firm 2 has no profitable deviation iff* $p_1 \leq 2.00$ *(and* $p_1 < 100$*).* If $p_1 \geq 2.01$ and $p_1 < 100$, matching p_1 gives it the half market at margin $p_1 - 2 > 0$. If $p_1 \leq 2.00$, any price $\leq p_1$ has margin ≤ 0 and any price above p_1 sells nothing: no gain. (If $p_1 \geq 100$, firm 1 earns 0 and gains by charging 50.50, so this is not an equilibrium either.)
- *Firm 1, with* $p_1 \leq 2.00$. It earns $\pi_1(p_1) = (p_1 - 1)(100 - p_1)$, which *increases* in p_1 up to 50.50. It would raise the price to $\min\{p_2 - \varepsilon, 50.50\}$ if that exceeds p_1, so it needs $p_2 - \varepsilon \leq p_1$: $p_2 = p_1 + \varepsilon$.
- *The ends.* If $p_1 \leq 1.00$, then $\pi_1 \leq 0$: for $p_1 < 1.00$ raising the price above p_2 gives 0, better; for $p_1 = 1.00$, $p_2 = 1.01$ and matching 1.01 pays $\frac{1}{2}(0.01)(98.99) > 0$, better. So $p_1 \geq 1.01$.
- *All these are equilibria.* For $1.01 \leq p_1 \leq 2.00$ and $p_2 = p_1 + \varepsilon$: firm 1 earns $\pi_1 > 0$; a lower price earns less (monotonicity), matching p_2 earns less (Lemma, because $p_2 \geq 1.02$), a higher price earns 0. Firm 2 has no profitable deviation because $p_1 \leq 2.00$.

(a) Besides $(1.99, 2.00)$: for instance $(1.01, 1.02)$, $(1.50, 1.51)$, $(2.00, 2.01)$.

(b) Exactly the profiles $(p_1, p_2) = (p_1, p_1 + 0.01)$ with $p_1 \in \{1.01, 1.02, \ldots, 2.00\}$.

Answer. (a) For example $(1.50, 1.51)$ or $(2.00, 2.01)$: firm 1 prices one cent below firm 2 and firm 2 earns zero but cannot profitably undercut. (b) There are exactly 100 pure-strategy Nash equilibria: $(p_1, p_1 + 0.01)$ for $p_1 = 1.01, 1.02, \ldots, 2.00$, which includes the book's $(1.99, 2.00)$.

(d) Interpretation. The low-cost firm is the only one that sells, and it sets the highest price that still undercuts the rival; the rival's price only needs to be one cent above, and the rival cannot match without losing money as long as $p_1 \leq 2.00 = c_2$. The range $[1.01, 2.00]$ is the set of prices between the two marginal costs that firm 1 can hold against the price-taking rival. With continuous prices the candidate limit $(2, 2)$ is not an equilibrium (Tadelis §5.2.4): there is no "one cent below" on a continuum, so firm 1 always wants to move closer to p_2. The grid model therefore has a multiplicity that the continuous model lacks; with symmetric costs on the grid, by contrast, Tadelis §5.2.4 finds just two equilibria, at c and at $c + \varepsilon$.

(e) Verification. `check_L2.py` computes the best-reply sets of both firms on the full grid $0.00, 0.01, \ldots, 100.00$ in integer arithmetic and intersects them: the equilibria are exactly the 100 pairs $(k, k+1)$ cents for $k = 101, \ldots, 200$. It also checks the book's $(1.99, 2.00)$ and $(2.00, 2.01)$, the non-equilibria $(1.00, 1.01)$ and $(2.01, 2.02)$, and the Lemma for every integer price from 2.00 to 99.99. Tadelis gives no answer.

(f) Common mistakes.

- Stating that $(1.99, 2.00)$ is the only equilibrium because "firm 2 charges its cost". Firm 2 can charge anything that leaves it with zero sales; equilibrium pins p_2 only through firm 1's best reply.
- Taking $p_1 = p_2 = 2.00$ as an equilibrium: firm 1 gains by undercutting by one cent.
- Forgetting the lower end: $(1.00, 1.01)$ is not an equilibrium, since firm 1 earns 0 and prefers to match 1.01.

Item 21: MWG Exercise 8.D.6

Advanced · difficulty 5/5

Statement (verbatim from the list, item 21)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 Exercises, Exercise 8.D.6ˆB, p. 264 [PDF p. 143]. Answer in the source: none in the book (the MWG solutions manual covers Chapter 8). Reproduced verbatim.

> Consider any two-player game of the following form (where letters indicate arbitrary payoffs):
>
> | Player 1 Player 2 | b_1 | b_2 |
> |---|---|---|
> | a_1 | u, v | ℓ, m |
> | a_2 | w, x | y, z |
>
> Show that a mixed strategy Nash equilibrium always exists in this game. [*Hint:* Define player 1's strategy to be his probability of choosing action a_1 and player 2's to be his probability of choosing b_1; then examine the best-response correspondences of the two players.]

(a) What is asked. Prove that every 2×2 game, with arbitrary payoffs, has a mixed-strategy Nash equilibrium.

(b) Method. The hint's route: describe each player's strategy by one probability, write the best-response correspondences, and show they always intersect (MWG §8.D, Def. 8.D.2 and Prop. 8.D.1, pp. 250–251, which is the slides' principle of indifference, Lecture 2, slides 44–46; Tadelis §6.4, pp. 121–122; Gibbons §1.3.B, pp. 41–44). Pure equilibria count: they are degenerate mixed strategies. Nash's theorem (slide 51) gives the result in general; the point here is a direct proof.

(c) Solution. Notation. $p = \sigma_1(a_1) \in [0,1]$, $q = \sigma_2(b_1) \in [0,1]$. Payoffs: $u_1(a_1,b_1) = u$, $u_1(a_1,b_2) = \ell$, $u_1(a_2,b_1) = w$, $u_1(a_2,b_2) = y$; $u_2(a_1,b_1) = v$, $u_2(a_1,b_2) = m$, $u_2(a_2,b_1) = x$, $u_2(a_2,b_2) = z$.

Step 1: the payoff gains are affine. Player 1's gain from a_1 over a_2 when player 2 plays q:

$$D_1(q) = [qu + (1-q)\ell] - [qw + (1-q)y] = q(u-w) + (1-q)(\ell - y). \tag{1}$$

Player 2's gain from b_1 over b_2 when player 1 plays p:

$$D_2(p) = [pv + (1-p)x] - [pm + (1-p)z] = p(v-m) + (1-p)(x-z). \tag{2}$$

Player 1's expected payoff is $u_1(p,q) = u_1(a_2,q) + p\,D_1(q)$, linear in p, so

$$BR_1(q) = \begin{cases} \{1\} & D_1(q) > 0, \\ [0,1] & D_1(q) = 0, \\ \{0\} & D_1(q) < 0, \end{cases} \qquad BR_2(p) = \begin{cases} \{1\} & D_2(p) > 0, \\ [0,1] & D_2(p) = 0, \\ \{0\} & D_2(p) < 0. \end{cases} \tag{3}$$

(p^*, q^*) is a NE iff $p^* \in BR_1(q^*)$ and $q^* \in BR_2(p^*)$.

Step 2: a sign lemma. An affine function D on $[0,1]$ satisfies exactly one of: (i) $D(t) \geq 0$ for all $t \in [0,1]$; (ii) $D(t) \leq 0$ for all t, with $D \not\equiv 0$; or (iii) $D(0)$ and $D(1)$ are nonzero with opposite signs. In case (iii) D has a unique zero $t^* = D(0)/(D(0) - D(1)) \in (0,1)$. (If $D(0)$, $D(1)$ are both ≥ 0 or both ≤ 0, then D, a convex combination of them, keeps that sign on $[0,1]$.)

Step 3: the proof, by cases on D_1.

Case A: $D_1(q) \geq 0$ for all q. Then $p = 1 \in BR_1(q)$ for every q by (3). Let $\hat{q} \in BR_2(1)$; it exists because (3) never gives an empty set. Then $1 \in BR_1(\hat{q})$ and $\hat{q} \in BR_2(1)$: $(1, \hat{q})$ is a NE.

Case B: $D_1(q) \leq 0$ for all q. Symmetrically, $p = 0$ is a best reply to every q, and $(0, \hat{q})$ with $\hat{q} \in BR_2(0)$ is a NE.

Case C: $D_1(0)$ and $D_1(1)$ have opposite signs. Player 1 is indifferent at exactly one $q^* \in (0,1)$. Solving $D_1(q^*) = 0$ from (1):

$$q^* = \frac{y - \ell}{(u - w) + (y - \ell)}. \tag{4}$$

Now look at D_2 and apply the lemma again.

- If $D_2(p) \geq 0$ for all p: $q = 1$ is a best reply to every p; with $\hat{p} \in BR_1(1)$, $(\hat{p}, 1)$ is a NE.
- If $D_2(p) \leq 0$ for all p: with $\hat{p} \in BR_1(0)$, $(\hat{p}, 0)$ is a NE.
- If $D_2(0)$, $D_2(1)$ have opposite signs: player 2 is indifferent at

$$p^* = \frac{z - x}{(v - m) + (z - x)} \in (0,1). \tag{5}$$

  At (p^*, q^*), $D_1(q^*) = 0$ gives $BR_1(q^*) = [0,1] \ni p^*$, and $D_2(p^*) = 0$ gives $BR_2(p^*) = [0,1] \ni q^*$. So (p^*, q^*) is a NE, in strictly mixed strategies.

The cases A, B, C exhaust all possibilities by the lemma, and each produces a NE. Hence every game of this form has a mixed-strategy Nash equilibrium. ■

(d) Interpretation. The argument is the 2×2 version of the fixed-point proof (Tadelis pp. 121–122): each best-response correspondence is a "step" from one side of the unit square to the other, with a vertical (or horizontal) segment where the player is indifferent, and two such steps must cross. Mixing is what makes this work: in pure strategies the steps have gaps (Matching Pennies has no pure NE), and the indifference segment closes the gap. Gibbons's four cases (p. 42) are the sign patterns of $D_1(0) = \ell - y$ and $D_1(1) = u - w$.

(e) Verification. `check_A2.py` checks (1), (2), (4) and (5) with sympy, then draws 20,000 random 2×2 games (half with integer payoffs in $\{-2, \ldots, 2\}$, so that ties and constant gains occur often), runs the case construction above, and confirms with the exact best-response test (3) that the profile it returns is a NE every time. *Comparison with the MWG solutions manual* (pp. 8-15–8-16, where the OCR in `Livros/` prints the number as "8.0.6"): it splits into the four strict-dominance cases ($u > w,\ \ell > y$; $u < w,\ \ell < y$; $v > m,\ x > z$; $v < m,\ x < z$) plus "all other cases", where it gives the mixed equilibrium $\alpha = (y - \ell)/(u + y - \ell - w)$ for player 2 and $\beta = (z - x)/(v + z - x - m)$ for player 1. These are our (4) and (5); the conclusion agrees. Two remarks on the manual: (i) in cases 3 and 4 it says "Player 2 will play his best response" where it means player 1 (a slip: the strategies named are a_1, a_2); (ii) its "all other cases" is not complete as written: if $u = w$ and $\ell = y$ (player 1 indifferent to everything) and player 2 has no strictly dominant strategy, the formula for α is $0/0$ (any α then works, so an equilibrium still exists, but the manual's formula does not exhibit it); the same holds for β when $v = m$ and $x = z$. The case split by the sign of D_1 above has no such gap.

(f) Common mistakes.

- Proving only the generic case. The formulas (4)–(5) divide by $(u-w)+(y-\ell)$ and $(v-m)+(z-x)$, which can be zero; cases A, B and the first two bullets of case C exist to cover exactly those games.
- Forgetting that a pure equilibrium is a mixed one. The claim is existence in $\Delta(S_1)\times\Delta(S_2)$; when one player has a dominant strategy the equilibrium found is pure, and that is enough.
- Citing Nash's theorem as the proof. It is true, but the exercise asks for a direct argument from the best-response correspondences, as the hint says.

Answer. With $p=\Pr(a_1)$, $q=\Pr(b_1)$, player 1's gain from a_1 is $D_1(q)=q(u-w)+(1-q)(\ell-y)$ and player 2's gain from b_1 is $D_2(p)=p(v-m)+(1-p)(x-z)$, both affine. If $D_1\geq 0$ (or ≤ 0) on $[0,1]$, player 1 has a pure strategy that is always a best reply, and pairing it with player 2's best reply to it is a NE; the same if D_2 keeps a sign. Otherwise both change sign, at $q^*=\frac{y-\ell}{(u-w)+(y-\ell)}$ and $p^*=\frac{z-x}{(v-m)+(z-x)}$ in $(0,1)$, and (p^*,q^*) is a NE because each player is indifferent. So a NE always exists.

Item 22: Tadelis Exercise 5.15

Advanced · difficulty 3/5

Statement (verbatim from the list, item 22)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 5.15 "Hotelling's Continuous Model", p. 99 [PDF p. 118]. Answer in the source: none (the preface, p. xiii, says about half of the exercises are solved on Princeton UP's website). Reproduced verbatim. **Reading, not covered in class** (Tadelis §5.2.5, pp. 93–95).

> **Hotelling's Continuous Model:** Consider Hotelling's model, in which citizens are a continuum of voters on the interval $A = [-a, a]$, with uniform distribution $U(a)$.
>
> a. What is the best response of candidate i if candidate j is choosing $a_j > 0$?
>
> b. Show that the unique Nash equilibrium is $a_1 = a_2 = 0$.
>
> c. Show that for a general distribution $F(\cdot)$ over $[-a, a]$ the unique Nash equilibrium is where each candidate chooses the policy associated with the median voter.

Referred to in the statement. Hotelling's model (Tadelis §5.2.5, pp. 92–94 [PDF p. 111–113]) (reproduced from the book):

> Two politicians, each caring only about being elected; each candidate i chooses a platform a_i (the policy of the citizen for whom it is ideal); each citizen votes for the candidate whose platform is closest to his preferences (an indifferent citizen tosses a coin); a candidate wins if a majority votes for him, and candidates prefer winning to a tie and a tie to losing.
>
> *Remark.* In the more common representation of the Hotelling competition model, the citizens are a continuum of voters, given by the interval $A = [\underline{a}, \overline{a}]$, with distribution $F(a)$ to determine the distribution of each political preference. We define the median voter as that voter a_m for which $\Pr\{a \leq a_m\} = \frac{1}{2}$. For example, if $F(\cdot)$ is uniform then $a_m = \frac{\underline{a}+\overline{a}}{2}$.

Optional extras.

- Tadelis 5.4 (p. 96, splitting pizza) and MWG 8.D.4 (p. 263, splitting 100 dollars, with dominance parts): more continua of equilibria, after item 4.

- Gibbons 1.13 (p. 51, two workers applying to two firms): pure and mixed equilibria in an applied 2×2 game, after item 8.

- Tadelis 6.6 (p. 124, declining industry): a 3×3 game where weak dominance shrinks the support of the unique mixed equilibrium.

- Muñoz-Garcia Ch. 2 Ex. 16 (p. 76, Cournot with efficiency changes) and Ch. 3 Ex. 10 (p. 114, rock-paper-scissors): worked answers in the book.

- Reading, not covered in class: Tadelis 5.7 (p. 96, the n-player commons), 4.8–4.9 (p. 78, the beauty contest); Gibbons 1.8 (p. 50, voting with two and three candidates); MWG 8.D.5 (pp. 263–264, ice-cream vendors).

(a) What is asked. Voters are a continuum on $A = [-a, a]$ with the uniform distribution. (a) What is the best response of candidate i to $a_j > 0$? (b) Show that the unique Nash equilibrium is $a_1 = a_2 = 0$. (c) Show that for a general distribution $F(\cdot)$ the unique Nash equilibrium is where both candidates choose the median voter's policy.

(b) Method. Best-response correspondences of a game with interval strategies, then the intersection (Lecture 2, slides 10 and 14). This is Hotelling's location game, in the voting version of Tadelis §5.2.5, pp. 92–94 (reading beyond the slides; the pharmacy location problem of Lecture 1 and Lecture 2, slides 2–3, is the same economics with discrete locations). A candidate prefers winning to a tie and a tie to losing; payoffs are 1, $\frac{1}{2}$, 0.

(c) Solution. Vote shares. A voter at x votes for the platform closest to x (a voter exactly in the middle splits evenly). If $a_1 < a_2$, candidate 1 gets the voters below the midpoint $m = \frac{1}{2}(a_1+a_2)$, share $F(m)$, and candidate 2 gets $1 - F(m)$; if $a_1 = a_2$ both get $\frac{1}{2}$. For the uniform distribution $F(x) = (x + a)/(2a)$, so $F(m) > \frac{1}{2}$ iff $m > 0$.

(a) Best response to $a_j > 0$.

- If $a_i < a_j$: candidate i is the lower platform, with share $F(m)$ and $m = \frac{1}{2}(a_i + a_j)$. He wins iff $m > 0$, i.e. $a_i > -a_j$, and ties iff $a_i = -a_j$.
- If $a_i = a_j$: tie.
- If $a_i > a_j$: he is the upper platform with share $1 - F(m)$, $m > 0$, so $1 - F(m) < \frac{1}{2}$: he loses.

Winning is best, and every platform in the open interval $(-a_j, a_j)$ wins, so

$$BR_i(a_j) = (-a_j, a_j) \quad (a_j > 0).$$

By symmetry $BR_i(a_j) = (a_j, -a_j)$ for $a_j < 0$, and $BR_i(0) = \{0\}$: against $a_j = 0$ any $a_i \neq 0$ loses and $a_i = 0$ ties. The endpoints $\pm a_j$ give a tie, strictly worse than winning, so they are not best replies.

(b) The equilibrium is $(0, 0)$. $0 \in BR_i(0)$ for both, so $(0, 0)$ is an equilibrium. Conversely, let (a_1, a_2) be an equilibrium. If $a_1 = 0$ then $a_2 \in BR_2(0) = \{0\}$. If both are non-zero, $a_1 \in BR_1(a_2)$ requires $|a_1| < |a_2|$ and $a_2 \in BR_2(a_1)$ requires $|a_2| < |a_1|$: impossible. If $a_2 = 0$, the same argument gives $a_1 = 0$. Hence $(0, 0)$ is the unique equilibrium.

(c) General F. Assume F is continuous and strictly increasing on $[-a, a]$, so that the median voter a_m ($F(a_m) = \frac{1}{2}$) is unique and $F(m) \gtreqless \frac{1}{2}$ iff $m \gtreqless a_m$. Take $a_1 < a_2$ with midpoint m: candidate 1 wins iff $m > a_m$, ties iff $m = a_m$, loses iff $m < a_m$.

- *(a_m, a_m) is an equilibrium.* A deviation to $x < a_m$ makes $m = \frac{1}{2}(x + a_m) < a_m$ and the deviator is the lower platform with share $F(m) < \frac{1}{2}$; a deviation to $x > a_m$ makes him the upper platform with share $1 - F(m) < \frac{1}{2}$. Either way he loses, worse than the tie.
- *Common platform $x \neq a_m$ is not an equilibrium.* If $x < a_m$, the rival moves to $x + \delta$ with δ small enough that $m = x + \delta/2 < a_m$; he is the upper platform with share $1 - F(m) > \frac{1}{2}$ and wins. For $x > a_m$ move below x.
- *Different platforms $a_1 < a_2$ are not an equilibrium.* If $m < a_m$, candidate 1 loses and gains by moving to a_2 (a tie); if $m > a_m$, candidate 2 loses and gains by moving to a_1; if $m = a_m$ (a tie), candidate 1 moves to $a_1 + \delta$, so the midpoint passes a_m and he wins.

So the unique equilibrium is $a_1 = a_2 = a_m$. For F uniform, $a_m = \frac{1}{2}(\underline{a} + \overline{a}) = 0$, which recovers (b). ■

Answer. (a) $BR_i(a_j) = (-a_j, a_j)$ for $a_j > 0$ (open interval, the platforms that beat a_j), $(a_j, -a_j)$ for $a_j < 0$, and $\{0\}$ for $a_j = 0$. (b) The only profile of mutual best replies is $a_1 = a_2 = 0$. (c) With F continuous and strictly increasing, the unique equilibrium is both candidates at the median voter a_m.

(d) Interpretation. This is the median voter theorem: competition for the majority pulls both platforms to the centre, whatever the shape of the distribution of preferences, and the equilibrium outcome is a tie. The best-response set is an open interval, not a point: any platform strictly inside the rival's mirror image wins, which is why $(0, 0)$ is the only crossing. The result is not covered in the slides (it is a reading item) but is the voting twin of Lecture 1's location game: with two competitors, location has a centre of gravity. Adding a third candidate changes the answer (Gibbons Problem 1.8).

(e) Verification. `check_L2.py` discretises the platforms (a grid of 0.1 on $[-1, 1]$, exact fractions, uniform voters) and finds all mutual best replies: the unique one is $(0, 0)$; it also checks that the best-response sets are the open intervals of part (a), the ties at $\pm a_j$ and the losses beyond. For a non-uniform F, namely $F(x) = 2x^2$ on $[0, \frac{1}{2}]$ and $1 - 2(1 - x)^2$ on $[\frac{1}{2}, 1]$ (continuous, strictly increasing, median $\frac{1}{2}$), platforms on a grid of $1/20$ give the unique equilibrium $(\frac{1}{2}, \frac{1}{2})$. These are checks of the continuum argument on a finite grid, not substitutes for it. Tadelis gives no answer.

(f) Common mistakes.

- Writing the best response as the single point $a_i = a_j - \varepsilon$ (a Bertrand-style undercut): here any platform in the open interval wins.
- Closing the interval: $a_i = \pm a_j$ is a tie, not a win.
- Skipping the uniqueness half of (b): $(0, 0)$ is shown to be an equilibrium, but one must also exclude profiles with different non-zero platforms.